



Stepwise benchmarking for multiple criteria sorting[☆]

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ABSTRACT

We propose a stepwise benchmarking framework for multiple criteria sorting, where alternatives are assigned to pre-defined and ordered decision classes. Our focus is on the alternatives currently attaining unfavorable classifications and aiming to reach a more preferred category through gradual performance modifications distributed over a few steps. We introduce strategies for the generation of development paths that incorporate either existing alternatives or fictive benchmarks. The latter ones are constructed using a suitably adapted framework for Post Factum Analysis. Its role is to highlight how the alternatives' performances need to be modified minimally so that the desired sorting recommendation is attained. The proposed method is applicable in the context of classifications arrived with precise values of preference model parameters or multiple feasible parameter sets within the framework of robustness analysis. A Decision Maker is allowed to specify the constraints on the feasible performance improvements and define whether to build the development path based on all criteria or a subset of criteria. We propose the measures that capture the balance in modifications of performances on different criteria and various steps. They can evaluate all generated improvement plans and indicate the path providing the smoothest development toward more preferred classes by following the intermediate benchmarks. We also offer some supportive measures that quantify the contribution of different criteria in attaining the target assignment. The use of the proposed framework is illustrated in a real-world problem of parametric evaluation of research units. We analyze the outcomes derived with a dedicated outranking-based approach employed by the Polish Ministry of Science and Higher Education and discuss the development plans for some example units assigned to the least preferred class.

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1. Introduction

Early research on benchmarking saw it as a systematic process for evaluating products, services, and work processes of an organization that are recognized as representing the best practices for organizational improvement [13,49]. Work on this topic conceptualized the term so that it can be applied in all organizations [6]. However, it was exclusively about searching for the best practices and performance improvements by "following the leader".

The beginning of the 21st century marks a new era in the use of benchmarking as a tool [3,5]. The Lisbon Treaty set goals for the development of the European Union (EU) countries in the decade to follow in a new form of policy coordination via the *Open Method of Coordination* (OMC) [42]. The OMC involves policy guidelines

with short, medium, and long-term goals; quantitative and qualitative indicators for benchmarking national performances against the best globally; period monitoring, evaluation, and peer review of member states. Such a document led to the extensive use of benchmarking in a policy context. It also pulled away from the concept of benchmarking from its original "business comparison roots" to a much broader set of public policy issues. The role of benchmarking did not lose its significance in the following years. The academic literature helped by polishing indicators and defining appropriate measures for specific policy goals. Europe 2020 strategy supported the claim. In particular, European Securities and Markets Authority (ESMA), i.e., the EU financial regulatory agency, has recently introduced benchmark regulations that ensure the accuracy and integrity of benchmarks and improve governance over the benchmark process. The United Nations (UN) followed the EU's concept and their 2030 Agenda for Sustainable Development with 17 Sustainable Development Goals (SDG) put the benchmarking process at the core of the future development for each country in the world.

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Parallel with the rise of benchmarking in a policy context, a new branch of benchmarking named *stepwise benchmarking* started to develop. The underlying idea is that often best practices are too advanced, and following the leader can be more of a burden than a way to improve. The proposed solution for performance improvement lies within gradual development by introducing *intermediate benchmarks*. The latter are treated as feasible practices that serve as a stepping stone of the development path to attain *ideal benchmarks* that are currently out of reach. The meaningfulness of such an approach was supported by a study revealing that setting the goals that were very challenging to attain rendered less than 50% of the targets actually achieved [33,38]. To the best of our knowledge, [20] were the first to coin the term of *stepwise path improvement*. It evolved to a stepwise benchmarking (SWBM), with its concept based on a set of intermediate benchmarks instead of a fixation on the leading performer [14,17,27,38–40].

The stepwise benchmarking process facilitates the knowledge absorption process and reduces the risk of failure implied by setting an out-of-reach target [13,17,24]. The SWBM, as well as the benchmarking concept itself, are mostly connected to Data Envelopment Analysis (DEA) [13,27,30,31,37], where one searches for the best practices on how to efficiently use the available inputs to transform them into produced outputs. When dealing with an inefficient Decision Making Unit (DMU), stepwise benchmarking was used to facilitate the improvement process [4,36,47]. Furthermore, SWBM found its use in ranking methods of Multiple Criteria Decision Analysis (MCDA) (see, e.g., [38,40]). An idea underlying the latter approaches is to generate the development paths by combining the alternatives from different layers in the preference hierarchy constructed with the outranking-based ranking approach. The example application of SWBM models concerned improvement of the performance of printer manufacturers [47], bank branches [4], or the EU countries in the field of digital connectivity [38,40].

The focus of this paper is on stepwise benchmarking in the context of multiple criteria sorting [51]. The latter's goal is to assign alternatives to pre-defined and ordered decision classes (categories). Such a scenario can be considered in many fields, where the final recommendation serves as a tool in policy or business context (see, e.g., ABC analysis [26], migration potential studies [2], or credit rating [12]). Although there are reliable, intuitive MCDA methods for supporting the Decision Makers (DMs) in deriving class assignments (see, e.g., [10,15,25,52]), less attention has been given to how to excel after the recommendation is obtained. This paper intends to fill the existing gap in the literature by offering a stepwise benchmarking model for sorting problems aimed to derive a development path for alternatives. In this way, we will provide tools for answering how to improve the current sorting classification and reach higher-level classes. In the basic scenario, we will consider the univocal results attained with precise parameter values incorporated into the sorting method. However, we will also generalize the framework to incorporate the robustness concern, taking into account the possible and necessary assignments computed with a set of sorting models compatible with the indirect or imprecise DM's preference information [19]. In the latter case, the desired class assignments need to be attained for at least one or all compatible sorting models.

We distinguish two types of methods for the path development within stepwise benchmarking literature – those that use existing alternatives as the benchmark targets (see, e.g., [4,13,36,40]), and those building development paths using non-existing alternatives (see, e.g., [30,31]). Our model will reconcile both approaches. The DM will be in a position to choose whether (s)he would like real or fictive alternatives as intermediate and ideal benchmarks. An idea underlying the generation of a development path for a given alternative will be to combine the alternatives satisfying requirements for being assigned to each class that is more preferred

than the currently attained one and at most as good as the desired class. By following such a scenario, the considered alternative can gradually improve its performances to reach some intermediate targets on the way to attaining the desired classification.

On the one hand, we will propose a path generation strategy using existing alternatives as points on the generated paths. Such real benchmarks are meaningful from a practical viewpoint because the path indicates from whom one should learn to attain the performance targets that have already proved to be realistic. On the other hand, we will outline the novel sorting-based stepwise benchmarking model based on the generation of fictive points. For this purpose, we will adapt Post Factum Analysis (PFA) [23], whose role is to highlight how the alternatives' performances need to be modified so that recommendation is changed in the desired way [8]. This will provide a setting for the development path's construction in a stepwise benchmarking manner. Its goal will be to construct the smoothest development toward a more preferred class by minimizing the required performance modifications when following its intermediate benchmarks.

Moreover, we will offer a mixed model where the DM can create a development path using feasible points but not necessarily coincident with the existing ones. For example, the DM would like to have an existing ideal benchmark, but (s)he is indifferent about the choice of intermediate benchmarks. This can be a realistic scenario for the cases where the method's goal is to enhance the learning process and reduce inefficiency on the route toward ideal goals [17].

To provide an interactive decision support system in which DM has as much control in the benchmarking process as possible, we allow him/her to decide whether (s)he wants to build the development path based on all criteria or a subset of criteria. This logic stems from public policy benchmarking issues. For example, suppose we discuss countries' development within SDG 2030 or Europe 2020 agenda. In that case, it is rational to assume a country would like to build capacities on a subset of specific indicators that it sees as the most important or the most realistic to improve. Moreover, in the method proposed in this paper, the DM is allowed to define constraints on the feasible values of performance improvements in each step.

Upon generating many development paths, a question arises on how to compare them and find the best development path in the stepwise benchmarking context. Paths within the literature are considered bad, good, useful, or smooth without a centralized definition of such concepts. The right development path should ensure a gradual improvement of an alternative en route to the top tier class, with as similar advancement within each step as possible [40]. We propose the measures that capture the balance in performance modifications on different criteria and various steps. Moreover, they admit the DM to define the importance of each criterion and step. This is essential in policy-based benchmarking as it provides an option for the DM to control dynamics within the benchmarking process. We also offer some supportive measures that quantify the contribution of different criteria in attaining the target assignment in terms of the magnitude of required improvements relative to the entire performance ranges. In this way, one can assess if all criteria equally contribute to the development of an alternative. Note that applying different measures provides richer information for the DM to opt for the most desired path according to her/his preferences.

The use of the proposed framework is illustrated on a real-world problem of parametric evaluation of research units [22]. The use of benchmarking methods in the field of education [48] and in the context of the evaluation of universities or research units has been of interest lately [9,46]. Such an evaluation is conducted every four years by the Polish Ministry of Science and Higher Education, resulting in units' assignment to four classes represent-

ing acquired effects and undertook activities. We analyze the outcomes derived by the Ministry with an outranking-based approach for a group of units representing exact and engineering sciences. We then show how the introduced method can be employed to generate the development plans for some example units assigned to the least preferred class. Moreover, we discuss the peculiarities of different proposed algorithms, the differences between performance measures, and the extensions of the basic approach.

The remainder of the paper is organized in the following way. Section 2 presents an outranking-based approach for parametric evaluation of research units. In Section 3, we introduce the algorithms for the generation of development paths based on the existing alternatives or fictive benchmarks generated with the evolutionary optimization methods. Section 4 discusses the measure for assessing the quality of improvement plans in terms of balance and contribution of various criteria. In Section 5, we report the results of a case study concerning the parametric evaluation of research units by providing example development paths. Section 6 discusses the usefulness and limitations of the proposed method. The last section concludes and provides avenues for future research.

2. Parametric evaluation of research units

Every four years, the Polish Ministry of Science and Higher Education performs a parametric evaluation of research units [22]. The evaluation is conducted individually for subsets of units representing the same scientific specialization, called a group of joint evaluation. A detailed description of the applied method can be found in Act no. 2154, published by the Ministry of Science and Higher Education.¹ In this section, we present basic notation and logic underlying the approach for parametric evaluation.

2.1. Notation and method

Let us use the following notation:

- $\mathcal{A} = \{a_1, \dots, a_i, \dots, a_n\}$ – a finite set of n alternatives;
- $G = \{g_1, \dots, g_j, \dots, g_m\}$ – a consistent family of m criteria [43], and $[g_1(a_i), \dots, g_j(a_i), \dots, g_m(a_i)]$ – a vector of performances of a_i ;
- $w = [w_1, \dots, w_j, \dots, w_m]$ – a vector of weights associated with the criteria; in the considered method, the weights are interpreted as importance coefficients;
- $\mathcal{W}$ – a space of feasible weights consistent with the DM's preferences translated into the linear constraints $A_w w \leq b_w$; when no preference information is available, $\mathcal{W} = \{w_j \geq 0, j = 1, \dots, m; \sum_{j=1}^m w_j = 1\}$;
- q_j and p_j – indifference and preference thresholds indicating, respectively, the maximal difference in performances that is negligible and the minimal differences that implies a strict preference; when a performance difference is greater than q_j and lesser than p_j , a weak preference relation holds on g_j ;
- $C = \{C_1, \dots, C_h, \dots, C_p\}$ – a set of p ordered categories, where C_1 and C_p are, respectively, the least and the most preferred ones;
- $T = \{t_1, \dots, t_h, \dots, t_{p-1}\}$ – a set of reference profiles separating the decision classes, expressed as the vectors of performances on all criteria, i.e., $t_h = [g_1(t_h), \dots, g_j(t_h), \dots, g_m(t_h)]$; t_h can be seen as a boundary between classes C_h and C_{h+1} ; moreover, $g_j(t_h) > g_j(t_{h-1})$ for $h = 2, \dots, p$ and $j = 1, \dots, m$.

In the considered problem of parametric evaluation of research units, the alternatives are compared against the boundary class profiles using an outranking-based preference model [44]. Let us

Table 1

Computation of the marginal preference index $\psi_j(a_i, t_h)$ for alternative a_i and profile t_h on criterion g_j .

Scenario	if $g_j(a_i) \geq g_j(t_h)$	if $g_j(a_i) < g_j(t_h)$
$\Delta_j(a_i, t_h) \geq p_j(a_i, t_h)$	1	-1
$p_j(a_i, t_h) > \Delta_j(a_i, t_h) \geq q_j(a_i, t_h)$	$\frac{(\Delta_j(a_i, t_h) - q_j(a_i, t_h))}{(p_j(a_i, t_h) - q_j(a_i, t_h))}$	$-\frac{(\Delta_j(a_i, t_h) - q_j(a_i, t_h))}{(p_j(a_i, t_h) - q_j(a_i, t_h))}$
$q_j(a_i, t_h) > \Delta_j(a_i, t_h)$	0	0

denote by $\Delta_j(a_i, t_h) = |g_j(a_i) - g_j(t_h)|$ the absolute value of difference between performances of alternative a_i and reference profile t_h on criterion g_j . Such a difference is compared against the indifference and preference thresholds [45] computed as follows (see Eqn 1 and 2):

$$q_j(a_i, t_h) = \alpha_{q_j} \cdot \min\{g_j(a_i), g_j(t_h)\} + \beta_{q_j}, \quad (1)$$

$$p_j(a_i, t_h) = \alpha_{p_j} \cdot \min\{g_j(a_i), g_j(t_h)\} + \beta_{p_j}, \quad (2)$$

where α_{q_j} , β_{q_j} , α_{p_j} , and β_{p_j} are the constant parameters of the affine functions defining the thresholds. The result is captured with a bipolar marginal preference index $\psi_j(a_i, t_h)$ (see Table 1). In particular, if a_i performs better than t_h by at least $p_j(a_i, t_h)$, then it is strictly preferred to t_h and hence $\psi_j(a_i, t_h) = 1$, whereas in case a_i is worse by at least $p_j(a_i, t_h)$, then $\psi_j(a_i, t_h) = -1$. However, when $\Delta_j(a_i, t_h)$ is not greater than $q_j(a_i, t_h)$, then $\psi_j(a_i, t_h) = 0$, indicating an indifference in the comparison of a_i and t_h .

When considering a weight vector $w \in \mathcal{W}$, the marginal preference indices are aggregated into the comprehensive preference index in the following way (see Eqn 3):

$$\Psi^w(a_i, t_h) = \sum_{j=1}^m w_j \cdot \psi_j(a_i, t_h). \quad (3)$$

When all criteria support a strict preference of a_i over t_h , then $\Psi^w(a_i, t_h) = 1$. In case t_h is more preferred on g_j , $j = 1, \dots, m$, then $\Psi^w(a_i, t_h) = -1$. Alternative a_i is assigned to the best class for which $\Psi^w(a_i, t_h) \geq 0$, indicating that a_i is at least as good as t_h given all criteria in G (see Eqn 4):

$$\begin{aligned} a_i &\rightarrow^w C_1 && \text{iff } \Psi^w(a_i, t_1) < 0; \\ a_i &\rightarrow^w C_h && \text{iff } \Psi^w(a_i, t_{h-1}) \geq 0 \text{ and } \Psi^w(a_i, t_h) < 0, \\ &&& \text{for } h = 2, \dots, p-1; \\ a_i &\rightarrow^w C_p && \text{iff } \Psi^w(a_i, t_{p-1}) \geq 0. \end{aligned} \quad (4)$$

When the analysis is performed for a precisely specified weight vector, each alternative is assigned to a single class only. However, when the results are investigated for space $\mathcal{W}$ of feasible weights, the assignments indicated for each alternative may differ from one weight vector to another. The diversity of outcomes that can be obtained for a_i is captured with:

- the possible assignment $C_p(a_i) = [C_p^L(a_i), C_p^R(a_i)]$, indicating the classes to which a_i is assigned for at least one $w \in \mathcal{W}$;
- the necessary assignment $C_N(a_i)$ indicating the class to which a_i is assigned for all $w \in \mathcal{W}$ (note that due to the applied assignment procedure, $C_N(a_i)$ is either a singleton or empty).

2.2. Case study: parametric evaluation of research units

In this paper, we will consider a group representing exact and engineering sciences. It is composed of 39 units, corresponding mainly to the faculties of computing, automatic control, electrical engineering, and electronics. We will refer to the units using their identifiers imposed by the Ministry (see Table 2).

The units are evaluated in terms of four criteria of gain type (greater performances are more preferred than smaller performances) with the following interpretation:

¹ Signed on December 27, 2016, available at <https://dziennikustaw.gov.pl/D2016000215401.pdf>.

Table 2

Performances and assignments of research units from the group representing exact and engineering sciences.

Unit id.	g_1	g_2	g_3	g_4	$C_p^L(a_i)$	$C_p^R(a_i)$	$C_M(a_i)$
2	45.93	2.00	0.00	17.00	C	C	C
4	64.37	973.93	2.02	92.00	B	A*	A
11	94.11	1121.00	5.06	92.00	A	A+	A+
36	51.88	235.20	0.68	35.00	C	B	B
389	62.12	210.06	1.23	46.00	B	B	B
390	70.25	194.05	0.81	58.00	B	B	B
395	90.54	242.61	0.68	63.00	A	A*	A
402	89.75	1165.43	4.34	83.00	A	A+	A+
403	63.08	480.46	6.11	58.00	B	A*	A
410	57.82	132.90	0.19	35.00	C	B	B
418	52.22	100.72	0.89	42.00	B	B	B
424	75.75	302.50	2.48	65.00	B	A	A
433	61.54	861.18	2.07	85.00	B	A*	B
442	57.45	498.39	1.62	43.00	B	B	B
448	79.03	269.77	7.48	80.00	A	A*	A
449	58.15	508.26	1.20	69.00	B	B	B
451	88.82	525.47	5.24	82.00	A	A*	A
458	53.72	284.18	0.86	50.00	B	B	B
463	70.27	1282.04	2.12	71.00	B	A*	A
466	60.86	323.00	2.92	75.00	B	B	B
476	53.66	203.09	0.45	38.00	B	B	B
485	77.09	1408.00	5.69	90.00	A	A*	A
486	63.76	524.51	3.02	41.00	B	B	B
503	80.18	1141.07	2.68	92.00	A	A*	A
504	112.37	330.90	4.76	88.00	A	A*	A
505	61.11	395.57	3.72	80.00	B	B	B
515	64.93	473.10	10.15	69.00	B	A*	A
529	60.29	11.00	0.00	0.00	C	B	B
825	62.5	108.56	4.68	40.00	B	B	B
897	65.65	345.43	2.34	76.00	B	B	B
904	27.58	14.00	4.17	33.00	C	C	C
910	77.24	678.00	13.32	48.00	A	A*	A
912	48.59	492.97	4.34	31.00	B	B	B
926	48.00	0.00	0.00	22.00	C	C	C
958	70.13	32.43	4.45	42.00	B	B	B
960	63.16	20.00	0.00	37.00	C	B	B
975	0.00	0.20	0.00	17.00	C	C	C
987	60.85	367.16	1.47	71.00	B	B	B
988	55.66	274.40	0.38	65.00	B	B	B

- *scientific activity* g_1 , capturing the journal papers, monographs, books, chapters, patents, and design; the performance on this criterion corresponds to an average number of points gained for publications or inventions by a single researcher of the unit;
- *scientific potential* g_2 , built on the number of academic degrees granted to the unit's researchers, mobility of unit's members, certification of laboratories, right to grant academic degrees, contribution to the development of scientific projects, memberships in international scientific organizations or editorial boards of journals; all achievements are scored and summed up to get an evaluation;
- *material effects of the unit's activities* g_3 , referring to the financial aspects of the ongoing projects and cooperation with the industry (the points are granted for, e.g., the unit's income from the invented technologies, implemented software, elaborated algorithms, or commercialization of the research results);
- *non-material effects of the unit's activities* g_4 – evaluation of ten most important achievements of unit's members (e.g., study results that have an impact on societies, scientific awards, excellence in research certificates, or organization of international conferences) conducted by experts.

A detailed procedure for computing the performances of research units on the four criteria can be found in the already mentioned Act no. 2154. We include the crucial points of this description in e-Appendix 1 (supplementary material available online).

For the performances of the considered units on all criteria, see Table 2. They differ vastly from one unit to another. On the one

hand, units 2 and 975 perform poorly in terms of all relevant characteristics. On the other hand, units 11, 402, 485, and 503 attain relatively high performances on all criteria. The greatest performances on the particular criteria are: 112.37 on g_1 , 1408.00 on g_2 , 13.32 on g_3 , 92.00 on g_4 . These are attained by units 4, 11, 485, 503, 504, and 910. However, there is no single unit that would dominate the remaining ones. The performances of all units are visualized in Fig. 1. We use a three-dimensional chart, where the axes correspond to criteria g_1 , g_2 , and g_3 . The performances on criterion g_4 are represented with a color on a pre-defined scale (the darker it is, the worse the performance on g_4).

The weights assigned to the criteria by the Ministry for the evaluation of units representing exact and engineering sciences are as follows: $w_1 = 0.65$, $w_2 = 0.10$, $w_3 = 0.15$, and $w_4 = 0.10$. Hence the greatest priority is associated with the results of scientific activity. However, to investigate the robustness of classification outcomes, we will also apply imprecise weights defined as intervals delimited by the extreme values used by the Ministry in the evaluation of groups representing other fields of science. Space $\mathcal{W}$ of feasible weights is defined as follows:

$$\mathcal{W} = \{(w_1, w_2, w_3, w_4) : w_1 + w_2 + w_3 + w_4 = 1,$$

$$w_1 \in [0.45, 0.7], w_2 \in [0.05, 0.15], w_3 \in [0.1, 0.25],$$

$$w_4 \in [0.05, 0.15]\}.$$

The parameters of the affine functions defining the indifference and preference thresholds are $\alpha_{q_j} = 0.1$, $\alpha_{p_j} = 0.3$, and $\beta_{q_j} = \beta_{p_j} = 0$.

The units are assigned to four preference ordered classes: C, B, A, and A+, where C and A+ are, respectively, the least and the most preferred ones. The categorization into classes C, B, and A is conducted according to the method described in Section 2.1. It exploits the results of comparisons of units with the reference profiles separating classes C and B (*JRB*) as well as classes B and A (*JRA*). For the considered group of units, these profiles were defined by the Ministry in the following way: $JRB = [40.57, 518.96, 3.06, 40.95]$ and $JRA = [63.10, 807.28, 4.76, 63.705]$ ² If the unit is classified as A, it can be viewed as a potential candidate for class A+, granted only to the most distinguishing units. The Ministry establishes the rules for each group of joint evaluation based on the analysis of performances attained by units assigned to class A. If some units prove to be clearly better than the vast majority of the remaining ones, they are assigned to the most preferred class. Therefore, when analyzing the results for some group of units, one can assume that to satisfy the conditions for such an assignment, a given unit should either dominate or attain a comprehensive preference degree at least as good as some unit classified as A+ by the Ministry.

The assignments C_M derived for 39 units by the Ministry are provided in Table 2. The cardinalities of all classes are as follows: C – 4, B – 21, A – 12, and A+ – 2. In particular, units 2, 904, 926, and 975 were assigned to the least preferred class, whereas units 11 and 402 were evaluated as the most advantageous. The least C_p^L and the most C_p^R preferred possible assignments derived from the analysis of interval weights are also provided in Table 2. Some units (e.g., 2 or 389) are univocally assigned to the same class by all feasible weights. For some other units, the recommendation differs from one weight vector to another. For example, unit 36 can possibly be assigned to class C or B. For the units that attained a score at least as good as one of the units originally classified as A+, we denoted the best possible class by A*. This means that they potentially attain the same comprehensive score as units assigned

² See <https://www.gov.pl/web/edukacja-i-nauka/wyniki-kompleksowej-oceny-jakosci-dzialalnosci-naukowej-lub-badawczo-rozwojowej-jednostek-naukowych-2017> for more details.

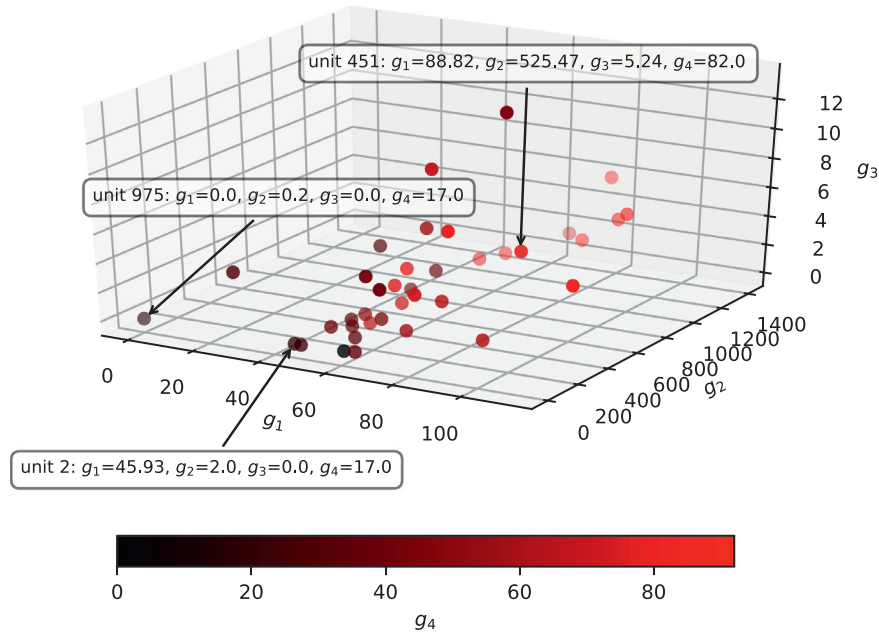


Fig. 1. Performances of 39 units on four criteria ($g_1 - g_4$) (three arbitrarily selected units (2, 451, and 975) are annotated).

to the most preferred class for at least one feasible weight vector. However, since they do not dominate any of these units, such an assignment would need to be confirmed by the Ministry. Clearly, for all units $a_i \in \mathcal{A}$, $C_M(a_i) \in [C_P^L(a_i), C_P^R(a_i)]$.

3. Algorithms for generation of development paths in multiple criteria sorting

The implementation of stepwise benchmarking methods begins upon deriving a recommendation from some sorting procedure. The first phase consists of the construction of development paths, i.e., possible ways to improve en route from the currently achieved class C_S to a desired one C_D for alternative $a_i \in \mathcal{A}$. Stepwise benchmarking sees development paths as advice on how to develop most efficiently and smoothly to successively attain the requirements for classes: C_{S+1} , C_{S+2} , ..., C_{D-1} , and C_D . Whichever the desired class C_D , its achievement can be analyzed given a single weight vector w or a set $\mathcal{W}$ of weight vectors. The former is useful, e.g., when dealing with the recommendation derived with a precise preference model. The latter can be employed when considering a DM whose preference model was elicited either indirectly or the admissible parameter values were delimited with some linear constraints. Furthermore, when accounting for multiple weight vectors in $\mathcal{W}$, the target achievement may be considered in the possible or necessary sense, i.e., for at least one or all feasible weights, respectively. We will use the following notation (see Eqn 5) concerning the targets:

$$T_Z^Y(a_i, C_S, C_D), \quad (5)$$

where:

- $Y \in \{w, \mathcal{W}\}$ represents either a single weight vector w or a set of feasible weight vectors $\mathcal{W}$;
- $Z \in \{P, N\}$ with P and N representing, respectively, the possibility or the necessity of achieving the assignment (note that in the case $Y = w$, Z can be neglected).

For example, $T^w(a_i, C_S, C_D)$ concerns a_i being currently assigned to C_S and aiming at being assigned to C_D in the classification imposed by w . In turn, $T_N^{\mathcal{W}}(a_i, C_S, C_D)$ is related to a_i , currently in class C_S , to be assigned to a class at least C_D for all feasible weights in $\mathcal{W}$.

We denote the achieved target by $T_Z^Y(a_i, C_S, C_D) = 1$, and we write $T_Z^Y(a_i, C_S, C_D) = 0$, otherwise.

Previous works saw authors opt either on real, existing alternatives or fictive benchmarks. We do not single out any option, leaving this decision to the DM having expertise and insight on the problem specifics. For this reason, we provide an overall decision support system for any preferred approach. We define three main options for generating development paths. The first, called *Real-Path*, uses existing alternatives as benchmarks on the paths. The remaining two strategies, called *EMO-Start-Path* and *EMO-Seq-Path*, require the generation of new, potentially fictive points that minimize the performance changes needed to attain some intermediate class on the way to reach the desired classification. For this purpose, the latter two methods incorporate the Evolutionary Multiple Objective Optimization (EMO) algorithm, called NSGA-II. Selection of EMO is motivated by the multiple objective characteristics of the tackled problem that becomes non-linear when both weights and performances are variables, as well as the intent of exploiting the multiplicity of possible solutions that will be generated in a single run.

All approaches generate subsets of real or fictive alternatives. If the considered alternative attains their performances, it will attain some class on a path from the already achieved to the desired one. Hence the development paths are ordered sequences of alternatives contained in the subsets of pre-generated options that allow achieving classes C_{S+1} , C_{S+2} , ..., C_{D-1} , and C_D . The set of all possible paths corresponding to attaining such a target will be denoted by $\Pi_Z^Y(a_i, C_S, C_D)$.

3.1. Real-Path: generation of development paths using existing alternatives

Path generation using existing alternatives is perceived as the most realistic in numerous scenarios (see [27] for more details). Real benchmarks are meaningful from a practical standpoint. Public policy benchmarking can serve as a perfect example. Such a strategy provides an opportunity for the DM to take an in-depth, qualitative look at intermediate benchmark actions – a foundation for future policy measures. Specifically, the development path indicates the alternatives that have already attained the desired tar-

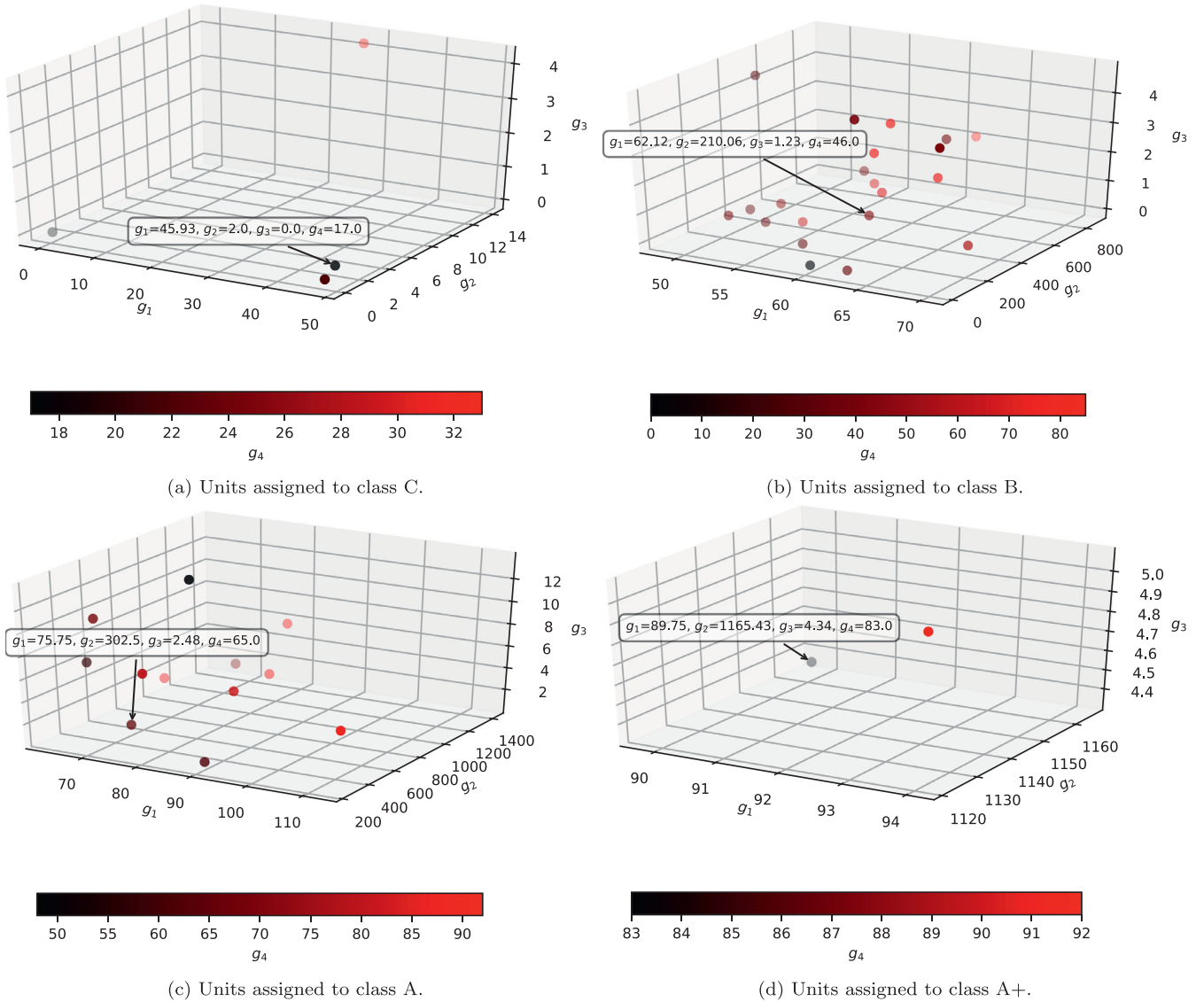


Fig. 2. All possible development paths generated with the *Real-Path* algorithm for unit 2 assigned to class C with the current performances.

gets, hence suggesting from whom one should learn. However, the drawback of such an approach derives from a bounded set of possible development paths, which can result in a lack of flexibility. The existing alternatives may not cover the whole space of solutions, neglecting some potential good options that are not present in the analyzed data set.

Let us denote by $\mathcal{A}(C_h) \in \mathcal{A}$ a subset of existing alternatives with performances that would allow a_i to be assigned to class C_h . In case the analysis is performed for a single weight vector w , these are alternatives precisely assigned to C_h ($C_M(a_i) = C_h$). When dealing with a feasible weight space $\mathcal{W}$, for the possible setting – we would refer to the alternatives with $C_P^R(a_i) = C_h$ and for the necessary setting – $C_P^L(a_i) = C_h$. Hence when considering robustness analysis, $\mathcal{A}(C_h)$ is formed by the alternatives assigned to C_h in the most or the least advantageous scenarios. Overall, all possible development paths are generated with the *Real-Path* framework in the following way (see Eqn 6):

$$\Pi_Z^Y(a_i, C_S, C_D) = a_i \times \mathcal{A}(C_{S+1}) \times \mathcal{A}(C_{S+2}) \times \dots \times \mathcal{A}(C_{D-1}) \times \mathcal{A}(C_D). \quad (6)$$

The development paths generated in this way for the considered problem of parametric evaluation of research units are presented

in Fig. 2. In particular, when considering unit 2, classified as C and marked in Fig. 2a, we may analyze 504 different paths by combining the units assigned to classes B, A, and A+ in various ways. We distinguish the path that allows unit 2 to satisfy the requirements for being classified as A+ by successively reaching the performances of units 389 (for class B), 424 (for class A), and 402 (for class A+). The latter path is also referred to in the following sections.

3.2. EMO-Path: generation of development paths using fictive benchmarks obtained with the evolutionary multiple objective optimization algorithm

The other two strategies require the generation of new, potentially fictive points when constructing the improvement paths, similarly to the approaches proposed in [14,30,31,50]. The underlying assumption is that development paths should minimize the performance changes on all criteria required to attain a more preferred class. Let us consider target $T_Z^Y(a_i, C_S, C_h)$, $S < h \leq D$, involving alternative $a_i \in \mathcal{A}$ from class C_S , aiming at being assigned to class C_h . Let $B(T_Z^Y(a_i, C_S, C_h))$ denote the set of all performance changes $[b_1^{i,h}, \dots, b_j^{i,h}, \dots, b_m^{i,h}]$ that, when applied to the

performances of a_i , allow attaining the target assignment, i.e., $T_Z^Y(a_i, C_S, C_h) = 1$. The problem that needs to be solved for computing such minimal performance changes can be formulated as follows (see Eqn 7):

$$\text{Minimize } b_1^{i,h}, \dots, b_j^{i,h}, \dots, b_m^{i,h}, \quad (7)$$

s.t.:

$$\left. \begin{array}{l} \text{for all } j = 1, \dots, m : \\ b_j^{i,\min} \leq b_j^{i,h} \leq b_j^{i,\max}, \\ g_j(a_i^{S,h}) = g_j(a_i) + b_j^{i,h}, \\ \text{if } Y = w : \\ w = [w_1, \dots, w_j, \dots, w_m]; \\ \Psi^w(a_i^{S,h}, t_{h-1}) = \sum_{j=1}^m w_j \cdot \psi_j(a_i^{S,h}, t_{h-1}) \geq 0, \\ \text{if } Y = \mathcal{W} : \\ \left. \begin{array}{l} w_j^{\min} \leq w_j \leq w_j^{\max}, \quad j = 1, \dots, m, \\ \sum_{j=1}^m w_j = 1, \end{array} \right\} \mathcal{W} \\ \text{if } Z = N, \text{ then for all } w \in \mathcal{W} : \\ \Psi^w(a_i^{S,h}, t_{h-1}) = \sum_{j=1}^m w_j \cdot \psi_j(a_i^{S,h}, t_{h-1}) \geq 0, \\ \text{if } Z = P, \text{ then for some } w \in \mathcal{W} : \\ \Psi^w(a_i^{S,h}, t_{h-1}) = \sum_{j=1}^m w_j \cdot \psi_j(a_i^{S,h}, t_{h-1}) \geq 0, \end{array} \right\} E(T_Z^Y(a_i, C_S, C_h), a_i^{S,h})$$

where a_i is an alternative whose performances are modified, $b_j^{i,\min}$ and $b_j^{i,\max}$ are pre-defined minimal and maximal allowed modifications on criterion g_j (by default, $b_j^{i,\min} = 0$ and $b_j^{i,\max} = \infty$), $w = [w_1, \dots, w_j, \dots, w_m]$ is a pre-defined vector of precise weights, and $\mathcal{W}$ is a set of linear constraints defining a feasible weight space (note that we refer to the types of constraints that are used in this paper (e.g., $w_j^{\min}$ and $w_j^{\max}$ are extreme allowed values of weight w_j ; however, other types of linear constraints can be also handled). Let us denote the subset of Pareto-optimal solutions for which $T_Z^Y(a_i, C_S, C_h) = 1$ as $B^{PF}(T_Z^Y(a_i, C_S, C_h)) \subseteq B(T_Z^Y(a_i, C_S, C_h))$. For each performance change vector in $B^{PF}(T_Z^Y(a_i, C_S, C_h))$ there is no other vector that would allow achieving the assignment to C_h with not greater changes on all criteria and strictly less change on at least one criterion [8]. The set of solutions obtained by applying these performance changes to a_i will be denoted by $\mathcal{A}(T_Z^Y(a_i, C_S, C_h))$ and an individual solution contained in this set by $a_i^{S,h}$.

To solve the above Multiple Objective Optimization (MOO) problem efficiently, we applied an evolutionary algorithm called NSGA-II [11]. The algorithm identifies a reliable approximation of $B^{PF}(T_Z^Y(a_i, C_S, C_h))$, i.e., non-dominated solutions for a given MOO problem. It starts with the initialization of a random population of solutions of size N . It then evolves the population in the subsequent generations by promoting solutions that are non-dominated and maintaining a uniform spread-out of the approximated Pareto front [11]. In this way, the evolutionary process generates a population of more favorable solutions than those contained in the preceding populations. The process is iterated until a stopping criterion is met. The latter is usually set to a pre-defined number of generations sufficient for the algorithm to converge to the Pareto front.

Based on the above general MOO for generating a set of fictive benchmarks $a_i^{S,h} \in \mathcal{A}(T_Z^Y(a_i, C_S, C_h))$ for class C_h , we propose a pair of strategies for generating the development paths allowing a_i to successively improve on the way from class C_S through $C_{S+1}, \dots, C_{D-1}$ to C_D . On the one hand, in *EMO-Start-Path*, we generate the subsets of performance changes and the respective new alternatives assigned to each class C_h , $h = S+1, \dots, D$, by modifying the considered alternative $a_i \in \mathcal{A}$. Hence, the computations for all classes are independent of each other because the same alternative is treated as a starting point in the context of all classes on

the development path. The more preferred the desired class, the greater the required performance changes. Overall, the possible development paths are generated by combining alternatives from the sets associated with the respective classes (see Eqn 8):

$$\Pi_Z^Y(a_i, C_S, C_D) = a_i \times \mathcal{A}(T_Z^Y(a_i, C_S, C_{S+1})) \times \mathcal{A}(T_Z^Y(a_i, C_S, C_{S+2})) \times \dots \times \mathcal{A}(T_Z^Y(a_i, C_S, C_{D-1})) \times \mathcal{A}(T_Z^Y(a_i, C_S, C_D)). \quad (8)$$

All possible development paths generated with the *EMO-Start-Path* algorithm for unit 2 are presented in Fig. 3. The considered unit is depicted in Fig. 3a. The following three sub-figures (b, c, and d) present the fictive units generated by applying to unit 2 the vectors of minimal performance modifications obtained with EMO, allowing to reach classes B, A, and A+, respectively. The paths are constructed by combining individual points from each sub-figure, i.e., joining a considered unit from class C (see Fig. 3a) to all fictive units in class B (see Fig. 3b) that are joined to all benchmarks in class A (see Fig. 3c), which are finally combined with all points satisfying the requirements of class A+ (see Fig. 3d). A single development path is distinguished in the figure by annotating the selected points with the respective performance vectors.

On the other hand, in the *EMO-Seq-Path* strategy, each fictive benchmark for an intermediate class in the development path is treated as an input for the next step. Hence, in the first step, existing alternative a_i assigned to class C_S is transformed, by means of minimal performance changes, into a set of alternatives $a_i^{S,S+1} \in \mathcal{A}(T_Z^Y(a_i, C_S, C_{S+1}))$ satisfying the requirements for being assigned to C_{S+1} and attaining the first intermediate target. Then, each fictive alternative $a_i^{S,S+1} \in \mathcal{A}(T_Z^Y(a_i, C_S, C_{S+1}))$ is transformed into a set of alternatives $a_i^{S+1,S+2} \in \mathcal{A}(T_Z^Y(a_i^{S,S+1}, C_{S+1}, C_{S+2}))$ assigned to class C_{S+2} , i.e., the second intermediate target. In the last step, all fictive alternatives $a_i^{D-2,D-1} \in \mathcal{A}(T_Z^Y(a_i, C_{D-2}, C_{D-1}))$ are used as starting points for generating a set of alternatives $a_i^{D-1,D} \in \mathcal{A}(T_Z^Y(a_i^{D-2,D-1}, C_{D-1}, C_D))$, satisfying the requirements for being assigned to the target class C_D . Thus, the computations performed in the current step depend on the results from the previous stage. For this reason, the optimization problems need to be solved sequentially. The development paths are generated by *EMO-Seq-Path* in the following way (see Eqn 9):

$$\Pi_Z^Y(a_i, C_S, C_D) = \prod_{a_i^{S,S+1} \in \mathcal{A}(T_Z^Y(a_i, C_S, C_{S+1}))} \prod_{a_i^{S+1,S+2} \in \mathcal{A}(T_Z^Y(a_i^{S,S+1}, C_{S+1}, C_{S+2}))} \dots \prod_{a_i^{D-2,D-1} \in \mathcal{A}(T_Z^Y(a_i^{D-3,D-2}, C_{D-2}, C_{D-1}))} \prod_{a_i^{D-1,D} \in \mathcal{A}(T_Z^Y(a_i^{D-2,D-1}, C_{D-1}, C_D))} [a_i, a_i^{S,S+1}, a_i^{S+1,S+2}, \dots, a_i^{D-2,D-1}, a_i^{D-1,D}]. \quad (9)$$

This process is visualized for the problem of parametric evaluation of research units in Fig. 4. The considered alternative (unit 2 assigned to class C) is presented in Fig. 4a. By solving a MOO problem, we are able to generate vectors of minimal performance changes on the four criteria that, when applied to unit 2, allow it to be assigned to class B. The fictive alternatives obtained in this way are shown in Fig. 4b. For each of these benchmarks, we compute the minimal modifications of performances that need to be applied for attaining class A. In Fig. 4c, we represent the fictive alternatives obtained in this way, when considering the alternative distinguished in Fig. 4b as a starting point. The analogous process is repeated in the final step (Fig. 4d depicts the fictive alternatives classified as A+ and obtained by modifying minimally an alternative distinguished in Fig. 4c). Hence, the structure of connections in the development paths can be visualized as a non-binary tree. Its root is in unit 2, which is connected to all fictive alternatives generated for class B and represented in Fig. 4b. Each of these points is connected to the new points constructed based on it using a plethora of minimal performance modifications for class A, etc. Overall, the only complete development paths that are repre-

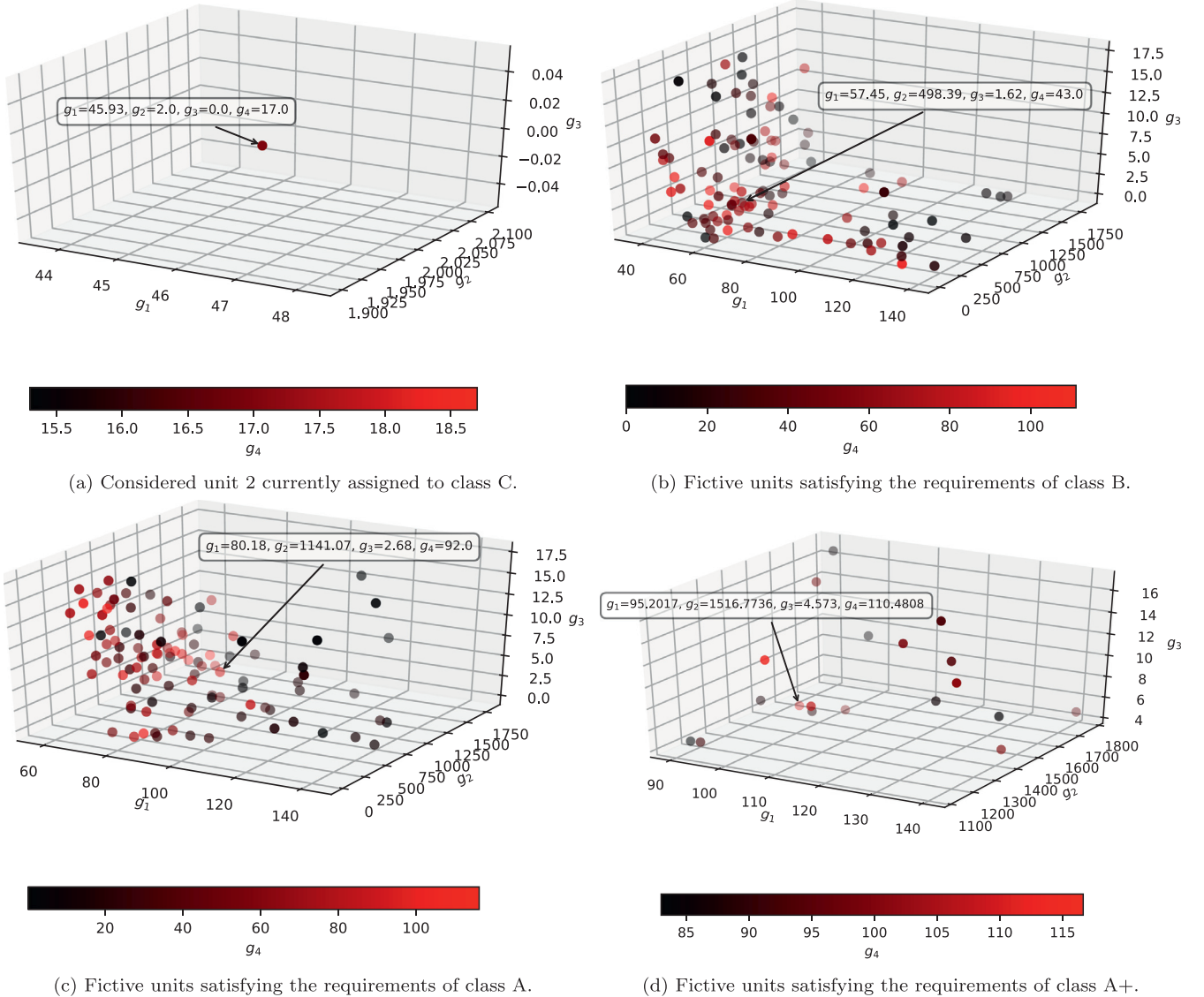


Fig. 3. All possible development paths generated with the *EMO-Start-Path* algorithm for unit 2 assigned to class C with the current performances (the exact performances of all units are provided in e-Appendix 2).

sented in Fig. 4 can be obtained by combining the three points distinguished in Fig. 4a, b, and c with one of the points illustrated in Fig. 4d. Moreover, all feasible paths start with unit 2 (Fig. 4a) that is combined with a single benchmark represented in Fig. 4b. Thus, Fig. 4 represents elements of all development paths, but complete paths shown in this figure constitute only a small subset of all paths generated by the *EMO-Seq-Path* strategy.

Note that the paths obtained with *EMO-Start-Path* and *EMO-Seq-Path* can be enriched with the plans built with the *Real-Path* strategy. This would ensure that the best development path constructed with *EMO-Path* would always be at least as good as all paths constructed with the use of existing alternatives. Moreover, the *EMO-Path* strategies can be customized to a particular decision problem. For example, when considering a parametric evaluation of research units, for the classes with pre-defined assignment conditions (C, B, and A), one can generate fictive benchmarks for attaining a given assignment with minimal performance changes. However, when the experts define the assignment conditions based on the analysis of performances attained by the existing alternatives, as for

class A+, it may be reasonable to consider the real options as the benchmarks.

The main benefit of the *EMO-Path* algorithms consists in generating a multiplicity of development paths, which are not influenced by the cardinalities of subsets of existing alternatives assigned to the respective classes. The variety of available options is important for enabling the DM to learn about the considered decision problem. The other advantage refers to the minimality of improvements needed to attain each intermediate target. The greatest disadvantage of the *EMO-Path* strategies, when compared to *Real-Path*, consists of high computation time. *EMO-Start-Path* needs to solve $D - S$ optimization problems, whereas for *EMO-Seq-Path* – the number of such problems in each step is N^{h-S+1} , for $h = S + 1, \dots, D$, where N is the population size. Finally, when the benchmarks in the generated paths do not correspond to the existing options, some of them may represent scenarios that cannot be realized in the real world. For this reason, the feasibility of outcomes derived from the *EMO-Path* strategies should be evaluated by the DMs.

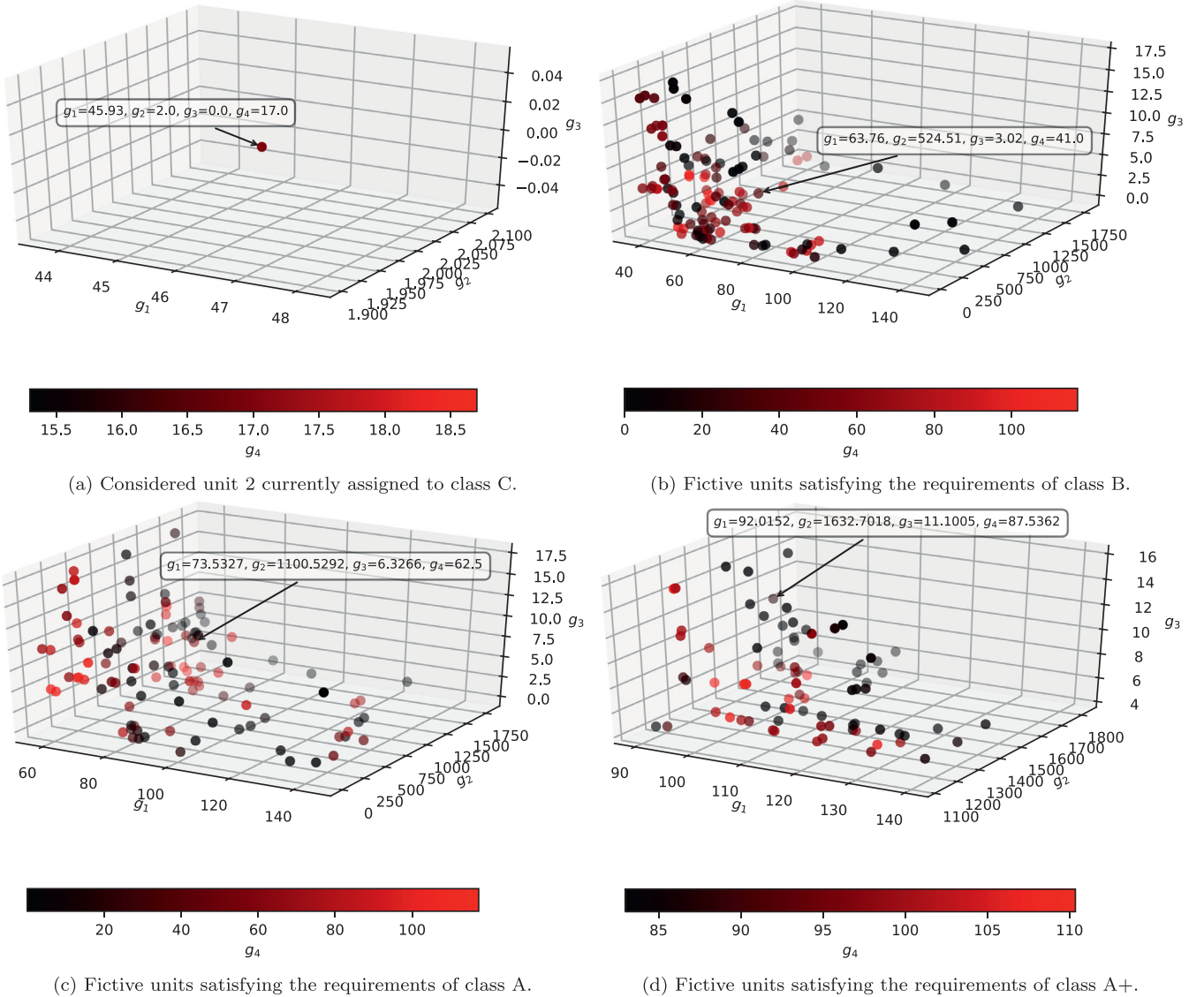


Fig. 4. Possible development paths generated with the *EMO-Seq-Path* algorithm for unit 2 assigned to class C with the current performances (the exact performances of all units are provided in e-Appendix 2).

4. Stepwise benchmarking measures for evaluation of the quality of development paths

In this section, we introduce a few measures that allow comparing the development paths and ultimately selecting the best one among multiple candidates. We first define a set of measures that refer to the evolution in progress, favoring simplicity and realism in implementing the proposed development paths in practice. This is because they promote smoothness of the development paths, i.e., evenly distributed performance improvements through all steps and on all relevant criteria [38]. For example, an ideally balanced improvement path from class C through B and A to class A+ would be the one that expects the unit to improve its performances by equal incremental steps in each stage (i.e., one-third of the overall performance gap on each criterion) as opposed to a plan assuming that the entire gap must be covered in a single step. We then discuss some supportive measures built on the absolute modification of performances. The measures represent different ideas for the stability of development, offering various perspectives on the quality of improvement scenarios. We will refer to the development path for alternative a_i currently assigned to class C_S and aiming at the classification to C_D in $|D - S|$ steps, denoted

by (see Eqn 10):

$$\pi(a_i, C_S, C_D) \in \Pi_Z^Y(a_i, C_S, C_D) = [a_i^S, a_i^{S+1}, \dots, a_i^{D-1}, a_i^D]. \quad (10)$$

Stepwise benchmarking methods see the optimal path as the one which offers the smoothest, most gradual development en route from the starting point to the target [38]. Let us first define a basic measure $BP(\pi(a_i, C_S, C_D))$ capturing the balance in performance modifications in all steps of path $\pi(a_i, C_S, C_D)$. Specifically, the measure is defined as a sum of differences between performance changes in each step and an average performance modification on all criteria (see Eqn 11):

$$\begin{aligned} BP(\pi(a_i, C_S, C_D)) &= \frac{1}{m \cdot |D - S|} \sum_{j=1}^m \sum_{h=S+1}^D bp_{j,h}(\pi(a_i, C_S, C_D)) \\ &= \frac{1}{m \cdot |D - S|} \sum_{j=1}^m \sum_{h=S+1}^D \left[\frac{(g_j(a_i^h) - g_j(a_i^{h-1})) - \mu_j(\pi(a_i, C_S, C_D))}{g_j^{\text{diff}}(\pi(a_i, C_S, C_D))} \right]^2, \end{aligned} \quad (11)$$

where $\mu_j(\pi(a_i, C_S, C_D)) = \frac{g_j(a_i^D) - g_j(a_i^S)}{D-S}$ is a gradual advancement within each step we are striving for and $bp_{j,h}(\pi(a_i, C_S, C_D)) = \left[\frac{(g_j(a_i^h) - g_j(a_i^{h-1})) - \mu_j(\pi(a_i, C_S, C_D))}{g_j^{\text{diff}}(\pi(a_i, C_S, C_D))} \right]^2$. Since the measure should be resistant to both positive and negative deviations, we use a second power of the difference. Moreover, to prevent the impact of different performance scales, for each criterion g_j the results are normalized by the comprehensive performance change imposed by the entire path on g_j , i.e., $g_j^{\text{diff}}(\pi(a_i, C_S, C_D)) = g_j(a_i^D) - g_j(a_i^S)$. When BP is low, the performance changes on all criteria are similar to each other, implying that the entire development path can be considered smooth. On the contrary, a high value of BP suggests significant differences in the modification expected from an alternative in various steps.

The BP measure can be generalized in two ways. On the one hand, we may consider the criteria weights w_j^p , $j = 1, \dots, m$, hence emphasizing the balance of performance changes on the more important criteria instead of promoting the paths that offer relatively smooth performance modifications on all criteria (see Eqn 12):

$$WBP(\pi(a_i, C_S, C_D)) = \frac{1}{|D-S|} \sum_{j=1}^m w_j^p \sum_{h=S+1}^D bp_{j,h}(\pi(a_i, C_S, C_D)), \quad (12)$$

where $\sum_{j=1}^m w_j^p = 1$. Note that weights w_j^p , $j = 1, \dots, m$, do not need to correspond to the importance coefficients w_j . In turn, they define the priorities for ensuring the balance of performance changes in the particular steps on the respective criteria.

On the other hand, we may account for the weights v_h^p , $h = S+1, \dots, D$ ($\sum_{h=S+1}^D v_h^p = 1$), associated with different steps in the development path. In some scenarios, the first step is the most important, e.g., in long-term public policy benchmarking processes, such as education reform, where political cycles do not coincide with project length. Then, the DM may wish to make the initial step as smooth as possible in the entire development of a given alternative and assign greater weight to it. On the contrary, the most critical step in some other problems can be the last one, where the desired target needs to be finally reached. In this perspective, the Stepwise BP measure is defined as follows (see Eqn 13):

$$SBP(\pi(a_i, C_S, C_D)) = \frac{1}{m} \sum_{j=1}^m \sum_{h=S+1}^D v_h^p \cdot bp_{j,h}(\pi(a_i, C_S, C_D)). \quad (13)$$

The most general measure capturing the smoothness of required performance changes, called Stepwise Weighted Balance, incorporates the weights associated with both criteria and steps (see Eqn 14):

$$SWBP(\pi(a_i, C_S, C_D)) = \sum_{j=1}^m w_j^p \sum_{h=S+1}^D v_h^p \cdot bp_{j,h}(\pi(a_i, C_S, C_D)). \quad (14)$$

The measures above interpret the balance of development paths in terms of a (weighted) average of deviations from smooth performance changes. In this way, more significant deviations from the gradual advancements on some criteria or steps can be compensated with other criteria or steps on which the required modifications are more smooth. To prevent such a compensation, we define yet another measure (MCP) that captures the maximal performance change across all criteria and steps, normalized by the comprehensive performance modification imposed by the entire path on g_j (see Eqn 15):

$$MCP(\pi(a_i, C_S, C_D)) = \max_{j=1, \dots, m; h=S+1, \dots, D} \left[\frac{(g_j(a_i^h) - g_j(a_i^{h-1}))}{g_j^{\text{diff}}(\pi(a_i, C_S, C_D))} \right]. \quad (15)$$

Different variants of BP or MCP can be used to select the best development path in terms of gradual improvement interpreted as the balance of performance modifications on all criteria in all steps. In what follows, we introduce a pair of measures interpreting the smoothness of development differently. Their role should not be to rank all possible paths from the best to the worst, but rather to evaluate a subset of the most advantageous paths in terms of BP or MCP to reveal some of their additional features and enrich the DM's knowledge. In this regard, they can be treated as supportive measures. Note that such a strategy of combining various measures, enriching the basic viewpoint with some additional perspective, proved useful, e.g., in the evaluation of interestingness of decision rules [16].

The supportive measures refer to the contribution of particular criteria in attaining the target. On the one hand, the DM may wish to reach the goal by modifying the performances on all criteria up to a similar level. This may happen, e.g., when an alternative performs poorly on all criteria, and it requires a balanced improvement in terms of all relevant characteristics to increase its rating. On the other hand, the DM may favor the development paths, modifying the performances on a small subset of criteria only. Such a requirement may be attributed to scenarios either when some alternative is poor only on some criteria or when considerable progress on some criteria is more realistic than on others. For this purpose, we need to collate the performance modifications imposed by the path with the performance ranges on the particular criteria. In this way, we can assess if the expected change is substantial or marginal. The RP measure that captures a perspective based on ranges of performances can be defined as follows (see Eqn 16):

$$RP(\pi(a_i, C_S, C_D)) = \frac{1}{m} \sum_{j=1}^m [r_j(\pi(a_i, C_S, C_D)) - \bar{r}(\pi(a_i, C_S, C_D))]^2, \quad (16)$$

where r_j states for the comprehensive modification of the performance on criterion g_j relative to the performance range on g_j delimited by the extreme performances observed in set of alternatives $\mathcal{A}$ (see Eqn 17):

$$r_j(\pi(a_i, C_S, C_D)) = \frac{\sum_{h=S+1}^D |g_j(a_i^h) - g_i(a_i^{h-1})|}{g_j^{\text{range}}}, \quad (17)$$

where $\bar{r}(\pi(a_i, C_S, C_D)) = \frac{1}{m} \sum_{j=1}^m r_j(\pi(a_i, C_S, C_D))$ reflects an average relative modification across all criteria, and $g_j^{\text{range}} = \max_{a_i \in \mathcal{A}} g_j(a_i) - \min_{a_i \in \mathcal{A}} g_j(a_i)$. The selection of a desired preference direction for the RP measure is left to the DM facing a particular decision problem. If (s)he opts for a path offering similar relative performance modification on all criteria, RP should be minimized. When the magnitudes of relative performance changes should differ vastly from one criterion to another, one should favor the path for which RP is high.

Similarly to the BP measure, RP can be generalized to incorporate the normalized criteria weights w_j^p , $j = 1, \dots, m$, specifying the importance degrees for the levels of expected relative performance changes on various criteria (see Eqn 18):

$$WRP(\pi(a_i, C_S, C_D)) = \sum_{j=1}^m w_j^p [r_j(\pi(a_i, C_S, C_D)) - \bar{r}(\pi(a_i, C_S, C_D))]^2. \quad (18)$$

Illustrative example. Let us consider an example development path $\pi_1(2, C, A+)$ for unit 2. It allows satisfying the requirements for the assignment to class $A+$ by successively attaining the performances of other existing units: 389, 424, and 402 assigned to

Table 3Development path $\pi_1(2, C, A+)$ for unit 2 based on the existing research units: 389, 424, and 402.

Units					Performance changes			Measure components					
g_j	2	389	424	402	$C \rightarrow B$	$B \rightarrow A$	$A \rightarrow A+$	g_j^{diff}	μ_j	g_j^{range}	$bp_{j,B}$	$bp_{j,A}$	$bp_{j,A+}$
g_1	45.93	62.12	75.75	89.75	16.19	13.63	14.00	43.82	14.61	112.37	0.0013	0.0005	0.0002
g_2	2.00	210.06	302.50	1165.43	208.06	92.44	862.93	1163.43	387.81	1408	0.0238	0.0644	0.1668
g_3	0.00	1.23	2.48	4.34	1.23	1.25	1.86	4.34	1.45	13.32	0.0024	0.0021	0.0091
g_4	17.00	46.00	65.00	83.00	29	19	18	66	22	92	0.0112	0.0021	0.0037

classes B, A, and A+, respectively (see Table 3 and Fig. 2). Thus the path refers to 4 criteria and 3 steps. The performance modifications and the components that can be used to compute the values of the proposed measures for all criteria and steps are given in Table 3.

Let us first illustrate the computation of the measures capturing the smoothness of the modification changes in $\pi_1(2, C, A+)$, while assuming the following criteria and step weights: $w_1^p = 0.65$, $w_2^p = 0.15$, $w_3^p = 0.1$, $w_4^p = 0.15$, $v_B^p = 0.25$, $v_A^p = 0.25$, and $v_{A+}^p = 0.5$. Hence, we assign the greatest importance to the first criterion and the last step. In the below computations, we explicitly refer to the first and second step on g_1 as well as to the third (last) step on g_2 and g_4 :

$$BP(\pi_1(2, C, A+)) = \frac{1}{4 \cdot 3} [(16.19 - 14.61)/43.82]^2$$

$$+ [(13.63 - 14.61)/43.82]^2 + \dots$$

$$+ [(862.93 - 387.81)/1163.43]^2 + \dots +$$

$$+ [(18 - 22)/66]^2 = \frac{1}{4 \cdot 3} [0.0013 + 0.0005$$

$$+ \dots + 0.1668 + \dots + 0.0037] = 0.0240;$$

$$WBP(\pi_1(2, C, A+))$$

$$= \frac{1}{3} \cdot 0.65 \cdot 0.0013 + \frac{1}{3} \cdot 0.65 \cdot 0.0005$$

$$+ \dots + \frac{1}{3} \cdot 0.15 \cdot 0.1668 + \dots + \frac{1}{3} \cdot 0.15 \cdot 0.0037 = 0.0145;$$

$$SBP(\pi_1(2, C, A+))$$

$$= 0.25 \cdot \frac{1}{4} \cdot 0.0013 + 0.25 \cdot \frac{1}{4} \cdot 0.0005$$

$$+ \dots + 0.25 \cdot \frac{1}{2} \cdot 0.1668 + \dots + 0.5 \cdot \frac{1}{2} \cdot 0.0037 = 0.0292;$$

$$SWBP(\pi_1(2, C, A+))$$

$$= 0.25 \cdot 0.65 \cdot 0.0013 + 0.25 \cdot 0.65 \cdot 0.0005$$

$$+ \dots + 0.5 \cdot 0.15 \cdot 0.1668 + \dots + 0.5 \cdot 0.15 \cdot 0.0037 = 0.0175;$$

$$MCP(\pi_1(2, C, A+))$$

$$= \max_{j=1,2,3,4; h=B,A,A+} \left[16.19/43.82, 13.63/43.82, \right.$$

$$\left. \dots, 862.93/1163.43, \dots, 18/66 \right] = 0.7417.$$

The values of the BP measures cannot be interpreted alone as they serve to compare multiple development paths, favoring these with lesser measure values. Nevertheless, let us note that since $WBP(\pi_1(2, C, A+)) < BP(\pi_1(2, C, A+))$ the path is more balanced on the more important criteria. Indeed, the required modifications on g_1 are very similar to each other in the three steps. Moreover, since $SBP(\pi_1(2, C, A+)) > BP(\pi_1(2, C, A+))$, the modifications in the most important, last step are far from the gradual advancement within each step. This is particularly evident for criteria g_2 and g_3 , on which the expected improvements are greater than in the previous two steps. For the same reason, the value of $MCP(\pi_1(2, C, A+))$ is very high. It indicates that the expected performance modification on g_2 in the last step is over 74% of the comprehensive improvement on this criterion to be made by unit 2.

When it comes to the RP measure, it captures the balance of performance modifications on all criteria relative to the performance range delimited by the extreme performances on a given criterion observed in the set of units:

$$RP(\pi_1(2, C, A+)) = \frac{1}{4} \left[\left(\frac{43.82}{112.37} - 0.5649 \right)^2 + \left(\frac{1163}{1408} - 0.5649 \right)^2 \right.$$

$$+ \left. \left(\frac{4.34}{13.32} - 0.5649 \right)^2 + \left(\frac{66}{1408} - 0.5649 \right)^2 \right]$$

$$= \frac{1}{4} [(0.3890 - 0.5649)^2 + (0.8263 - 0.5649)^2$$

$$+ (0.3258 - 0.5649)^2 + (0.7174 - 0.5649)^2]$$

$$= 0.0448.$$

5. Stepwise benchmarking applied to parametric evaluation of research units

This section reports the results for a case study concerning the parametric evaluation of research units. The outcomes discussed in Section 2.2 serve as the starting point for the analysis. In what follows, we compare the paths obtained with various methods and paths optimizing different measures. We then discuss some extensions of the basic approach that exploit a set of feasible criteria weights or use only a subset of criteria to attain some target. All computations were performed on the Intel Core I7-4790 CPU 3.6GHz and 32Gb RAM.

5.1. Comparison of results obtained with different algorithms for generation of development paths

In this section, we compare the algorithms for the generation of development paths proposed in Section 3. We refer to the strategies for path construction, the quality of the most advantageous paths captured by the BP measure, and computation time, without discussing the obtained paths in detail. We focus on the analysis of unit 2, which corresponds to the Faculty of Computing in Warsaw Vistula University. When considering a precise weight vector applied by the Ministry, this unit is assigned to class C. We will construct the long-term development paths that would admit the assignment to class A+ with classes B and A as the intermediate targets. We will refer to five algorithms: *Real-Path* as well as *EMO-Start-Path* and *EMO-Seq-Path* in two variants, either allowing the generation of fictive benchmarks for class A+ or restricted to the use of existing units for the most preferred class.

Using the *Real-Path* strategy, we generated 504 development paths by combining 21, 12, and 2 units assigned to classes B, A, and A+, respectively. These paths are visualized in Fig. 2. Since no additional computations were performed, the generation time was negligible. Most of the best paths in terms of BP involve unit 424 as an intermediate target to reach class A, whereas unit 910 is often used as a benchmark for this class in the worst paths. It is more challenging to indicate such trends for the remaining two classes: B and A+. In particular, units 11 and 402 occur with a similar frequency in the best paths. This is confirmed by the development paths that proved to be the best in terms of BP (see Table 4). They

Table 4The two best development paths in terms of the BP measure constructed by *Real-Path* for unit 2.

Path	g_j	C	B	A	A+	BP(π)	Path	g_j	C	B	A	A+	BP(π)
π_1^{Real}	g_1	45.93	62.12	75.75	89.75	0.0240	π_2^{Real}	g_1	45.93	62.12	75.75	94.11	0.0251
	g_2	2.00	210.06	302.50	1165.43			g_2	2.00	210.06	302.50	1121.00	
	g_3	0.00	1.23	2.48	4.34			g_3	0.00	1.23	2.48	5.06	
	g_4	17.00	46.00	65.00	83.00			g_4	17.00	46.00	65.00	92.00	

Table 5The two best development paths in terms of the BP measure constructed by each variant of *EMO-Start-Path* for unit 2.

Path	g_j	C	B	A	A+	$BP(\pi)$	Path	g_j	C	B	A	A+	$BP(\pi)$
The best development paths generated by $EMO\text{-}Start\text{-}Path$													
$\pi_1^{EMO\text{-}Start}$	g_1	45.93	57.45	80.18	95.20	0.0098	$\pi_2^{EMO\text{-}Start}$	g_1	45.93	58.15	80.18	95.20	0.0102
	g_2	2.00	498.39	1141.07	1516.77			g_2	2.00	508.26	1141.07	1516.77	
	g_3	0.00	1.62	2.68	4.57			g_3	0.00	1.20	2.68	4.57	
	g_4	17.00	43.00	92.00	110.48			g_4	17.00	69.00	92.00	110.48	
The best development paths generated by $EMO\text{-}Start\text{-}Path$ with A+ based only on the existing units													
$\pi_{1,Real}^{EMO\text{-}Start}$	g_1	45.93	62.44	75.89	94.11	0.0131	$\pi_{2,Real}^{EMO\text{-}Start}$	g_1	45.93	57.45	75.89	94.11	0.0141
	g_2	2.00	433.42	827.63	1121.00			g_2	2.00	498.39	827.63	1121.00	
	g_3	0.00	0.98	3.22	5.06			g_3	0.00	1.62	3.22	5.06	
	g_4	17.00	55.77	87.05	92.00			g_4	17.00	43.00	87.05	92.00	

can be seen as sequences of units: 2-389-424-402 (π_1^{Real}) and 2-389-424-11 (π_2^{Real}). In addition, path π_1^{Real} is illustrated in Fig. 2. Both development paths direct toward institutions that are valid existing benchmarks. Their work should be closely monitored and evaluated to reach meaningful conclusions for improvement. Unit 389, as a first intermediate benchmark for class B, should be the starting point of the analysis. It represents medium performances for this class. In turn, the unit chosen for class A attains poor performances given evaluations of all 12 units evaluated as A. Since units 402 and 11 assigned to class A+ are rather similar to each other, their impact on the path quality is small. Paths π_1^{Real} and π_2^{Real} differ only in terms of the benchmark unit selected for A+ and the respective BP values are very similar ($BP(\pi_1^{Real}) = 0.0240$ and $BP(\pi_2^{Real}) = 0.0251$).

When employing the two variants of *EMO-Start-Path*, the NSGA-II algorithm was parameterized with a population size of 1000 and a maximal number of generations of 1000. Moreover, the existing units for each class were added to the fictive ones obtained with the evolutionary method to increase the number of possible combinations. The *EMO-Start-Path* strategy admitting fictive alternatives in all steps generated 207480 paths in 3.58s (see Fig. 3), whereas the framework restricted to use the existing units for class A+ constructed 20196 paths in 2.54s. Due to the incorporation of fictive benchmarks, the number of generated paths is significantly greater than in the case of *Real-Path*. Such a variety of possible options implies that the best paths in terms of BP are more advantageous than those involving only the existing research units.

In Table 5, we present a pair of the best paths obtained with each variant of *EMO-Start-Path*. When it comes to $\pi_1^{EMO-Start}$ (see Fig. 3; BP = 0.0098) and $\pi_2^{EMO-Start}$ (BP = 0.0102) involving only fictive alternatives, their smoothness is about 2.5 times better than for π_1^{Real} and π_2^{Real} . In turn, restricting *EMO-Start-Path* to the use of only existing units as benchmarks for class A+ made the results slightly worse ($BP(\pi_{1,Real}^{EMO-Start}) = 0.0131$ and $BP(\pi_{2,Real}^{EMO-Start}) = 0.0141$), though still significantly better than when using only the existing units for all steps. This observation confirms that using fictive alternatives as the intermediate benchmarks may lead to more smooth development paths even if the ideal point corresponds to the existing entity.

As far as the use of the two variants of *EMO-Seq-Path* is concerned, the population size was set to a smaller value of 100 because the evolutionary algorithm is run individually for each point generated in a given step. Nevertheless, the computation

time was significantly higher than for *EMO-Start-Path*. The *EMO-Seq-Path* method admitting fictive alternatives in all steps generated 858238 paths in over 8690s (see Fig. 4 for a subset of these paths), whereas the strategy restricted to use the existing units for class A+ constructed 27036 paths in 153.9s.

In Table 6, we present a pair of the best paths in terms of BP obtained with the two variants of *EMO-Seq-Path*. Sequential generation of fictive points in all steps – minimizing the required performance modification with respect to the starting point from the previous step – led to slightly better outcomes than in the case of *EMO-Seq-Path*. Specifically, the $\pi_1^{EMO-Seq}$ path represented in Fig. 4 consists of smooth changes of performances on four criteria in the three steps. Surprisingly, both $\pi_1^{EMO-Seq}$ and $\pi_2^{EMO-Seq}$ incorporated an existing unit (486) as an intermediate benchmark for class B. When the benchmark for class A+ was selected among the real alternatives, the best paths (see $\pi_{1,Real}^{EMO-Seq}$ and $\pi_{2,Real}^{EMO-Seq}$ in Table 6) proved to be slightly worse than for the counterpart variant of the *EMO-Start-Path* strategy.

Overall, the computation time of *Real-Path* was negligible, for *EMO-Start-Path* – it was still low, whereas *EMO-Seq-Path* – it was very high. This observation is related to the number of generated development paths. It was a few hundreds when using only the existing units while running into tens or hundreds of thousands when admitting the non-existing benchmarks' construction. Such flexibility allows obtaining a more smooth strategy in terms of the required performance modifications in all steps and criteria. In the following experiments, we will employ *EMO-Start-Path* as an algorithm offering a high quality of generated development paths in a time not exceeding a few seconds.

5.2. The best development paths in terms of different measures

In this section, we will present the best development paths in terms of different measures introduced in Section 4. Similar to the previous section, we focus on the strategy that unit 2 should implement to attain class A+. To determine a variety of paths, we use *EMO-Start-Path* incorporating NSGA-II, using a greater population size of 5000 and a maximal number of generations of 1000. In this way, we generated 5683093 paths. They served as the basis for the selection of the best strategies according to various measures. For the measures incorporating weights, we will use the following values:

Table 6

The two best development paths in terms of the BP measure constructed by each variant of EMO-Seq-Path for unit 2.

Path	g_j	C	B	A	A+	$BP(\pi)$	Path	g_j	C	B	A	A+	$BP(\pi)$
The best development paths generated by <i>EMO-Seq-Path</i>													
$\pi_1^{EMO-Seq}$	g_1	45.93	63.76	73.53	92.01	0.0032	$\pi_2^{EMO-Seq}$	g_1	45.93	63.76	73.53	97.30	0.0036
	g_2	2.00	524.51	1100.53	1632.70			g_2	2.00	524.51	1100.53	1551.72	
	g_3	0.00	3.02	6.33	11.10			g_3	0.00	3.02	6.33	9.87	
	g_4	17.00	41.00	62.50	87.54			g_4	17.00	41.00	62.50	83.11	
The best development paths generated by <i>EMO-Seq-Path</i> with A+ based only on the existing units													
$\pi_{1,Real}^{EMO-Seq}$	g_1	45.93	62.12	75.75	89.75	0.0240	$\pi_{2,Real}^{EMO-Seq}$	g_1	45.93	62.12	75.75	94.11	0.0254
	g_2	2.00	210.06	302.50	1165.43			g_2	2.00	210.06	302.50	1121.00	
	g_3	0.00	1.23	2.48	4.34			g_3	0.00	1.23	2.48	5.06	
	g_4	17.00	46.00	65.00	83.00			g_4	17.00	46.00	65.00	92.00	

Table 7The best development path $\pi_{BP}(2, C, A+)$ in terms of BP and SBP for unit 2 aiming to attain class A+ generated with the EMO-Start-Path algorithm.

Benchmarks					Performance changes			Measure components					
g_j	C	B	A	A+	$C \rightarrow B$	$B \rightarrow A$	$A \rightarrow A+$	g_j^{diff}	μ_j	g_j^{range}	$bp_{j,B}$	$bp_{j,A}$	$bp_{j,A+}$
g_1	45.93	57.45	72.60	96.26	11.52	15.15	23.66	50.33	16.78	112.37	0.0109	0.0010	0.0187
g_2	2.00	498.39	853.12	1242.50	496.39	354.73	389.38	1240.50	413.50	1408	0.0045	0.0022	0.0004
g_3	0.00	1.62	2.43	4.55	1.62	0.81	2.11	4.55	1.52	13.32	0.0005	0.0240	0.0174
g_4	17.00	43.00	74.43	110.01	26.00	31.43	35.58	93.01	31.00	92	0.0029	0.0000	0.0024

- for the four criteria: $w_1^p = 0.65$, $w_2^p = 0.15$, $w_3^p = 0.1$, and $w_4^p = 0.15$, used in WBP and SWBP;
- for the three steps: $\nu_B^p = 0.25$, $\nu_A^p = 0.25$, and $\nu_{A+}^p = 0.5$, used in SBP and SWBP.

To save space, we account for the two measures in the main paper, whereas the remaining results are presented in e-Appendix 1.

The best development path $\pi_{BP}(2, C, A+)$ in terms of BP and SBP is presented in Table 7 and Fig. 5. The measure values are $BP = 0.0071$ and $SBP = 0.0077$. The path expects unit 2 to improve in the three steps by 50.33 on g_1 , 1240.50 on g_2 , 4.55 on g_3 , and 93.01 on g_4 . This means that, e.g., an average number of points attained by the unit's researchers for his/her publications or patents needs to be increased by over 50 in the next evaluation period (see g_1), whereas the scoring of scientific achievements of all researchers needs to be improved substantially from 2 to 1242.5. When considering the performance ranges on the four criteria, the greatest performance modifications are expected on g_2 and g_4 , and the least improvement is projected for g_3 .

High evaluation of $\pi_{BP}(2, C, A+)$ in terms of BP is implied by the relatively balanced improvements expected from unit 2 in the three steps on all criteria. For example, the expected changes on g_1 in different steps range from 11.52 to 23.66 with an average step of 16.78, whereas the modifications on g_2 range from 354.73 to 496.39 with an average step of 413.50. The most balanced steps regarding the expected improvements being the closest to the respective averages are on g_4 in the second step and on g_2 in the last step. On the contrary, the least balanced steps are on g_3 in the second stage and on g_1 in the last stage.

Note that this path was the best also in terms of SBP, which incorporates the weights for various steps. These weights were set to assign a greater priority to the balanced improvement in the last step ($\nu_{A+}^p > \nu_B^p = \nu_A^p$). The underlying motivation was built on two facts. If the expected final development stage is too small, one can ask if it was ever needed. If the improvements expected in the last step are too large, this can destroy the entire development process. From this perspective, it is reasonable to expect that the projected modifications when moving from class A to A+ are closer to the average ones. When it comes to path $\pi_{BP}(2, C, A+)$, the improvement on g_2 in the last step is very close to the average, whereas on g_1 , g_3 , and g_4 – it is the greatest among all three steps, though still not far from the average modifications expected from unit 2 on these criteria in the respective stages.

The best development paths in terms of WBP, SWBP, and MCP are presented in e-Appendix 1. In addition, the supplementary material provides the ranks of all paths that are the best given each measure according to all five measures. Moreover, to enrich the primary analysis, we illustrate the use of some supportive measures.

5.3. Development paths for attaining the necessary and possible targets

In this section, we incorporate the robustness concern into stepwise benchmarking. We focus on the analysis of results derived with robustness analysis presented in Table 2 for the interval criteria weights. These outcomes reveal a potential variety of class assignments, depending on selecting a particular weight vector. For illustrative purposes, we refer to unit 975, standing for the Faculty of Computer Science at the University of Tourism and Ecology in Sucha Beskidzka. It attains poor performances on all four criteria, being assigned to the least preferred class with all feasible weights. In this perspective, it is reasonable to design a long-term plan with class B as an intermediate goal and class A as the target assignment, while restraining from aiming at class A+ as this goal is too unrealistic. We consider the possible and necessary settings, where the projected benchmarks should allow for attaining the respective assignments for at least one or all possible weight vectors, respectively.

The benchmarks generated with EMO-Start-Path for the possible and necessary assignments to classes B and A are illustrated in Fig. 6. Note that the development paths are obtained by combining the points from sub-figures a) and c) for the possible setting or b) and d) for the necessary scenario. The distribution of the generated benchmarks for the two classes is very similar (compare, e.g., Fig. 6a and c). Their analysis reveals some interesting insights on the coalitions of criteria on which the performances need to be improved to attain a given target. For example, let us consider the possible assignment of unit 975 to class B (see Fig. 6a). When the performance on g_1 attains the level allowing to compare positively with the profile separating classes C and B, the improvements required on the remaining three criteria are small or none. For example, when a medium performance is attained at g_4 , the performances on g_2 and g_3 can remain very poor. When either of them reaches a higher level, the modification on g_4 can be decreased. On the contrary, when the modification on g_1 is too small, the remain-

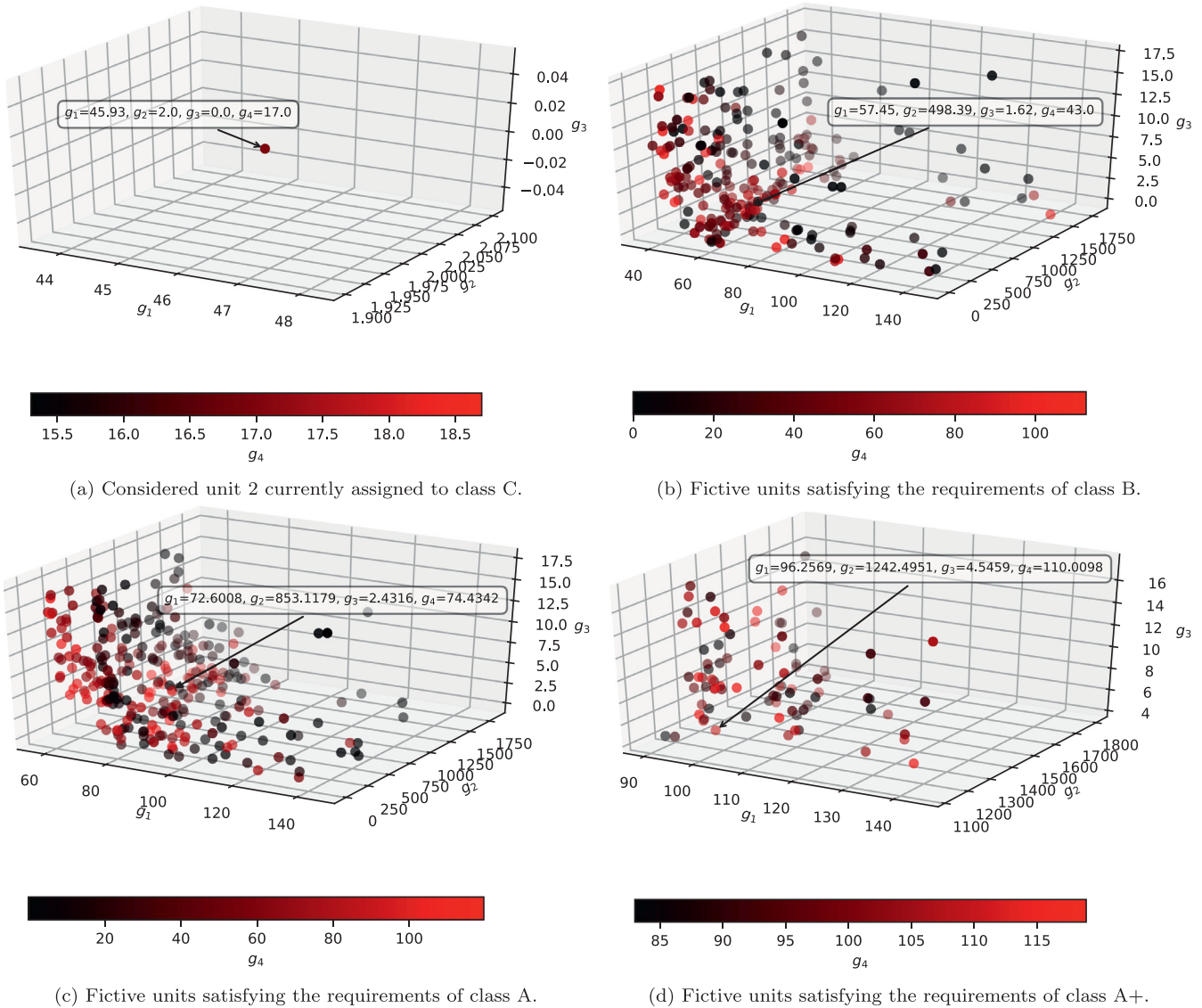


Fig. 5. The best development path $\pi_{BP}(2, C, A+)$ in terms of BP and SBP measures generated with the EMO -Start-Path algorithm for unit 2 assigned to class C with the current performances (the exact performances of all units are provided in e-Appendix 2).

ing three criteria need to be improved significantly. Similar observations can be formulated in the context of possible assignment to class A (see Fig. 6c). However, in this case, the required performance modifications on the respective criteria are greater because the assignment depends on how the unit compares with a better profile separating classes B and A.

When it comes to the necessary assignments, two substantial differences to the possible scenario can be observed. First, the required performance modifications are more significant. In particular, the improvement on the most important criterion g_1 needs to be substantial even for the intermediate target corresponding to class B. Second, the analysis of trade-offs between performance changes on various criteria reveals an even greater variety of development strategies. While the benchmarks depicted for the possible setting in Fig. 6a and c are clustered in two groups conditioned by the change on g_1 , there are four such groups in Fig. 6b and d. A general trend that can be observed in these figures is that a more significant improvement on g_2 admits lesser performance modification on g_4 , and vice versa.

Table 8

The best development paths in terms of BP measure generated for unit 975 currently assigned to class C in view of the necessary and possible assignments to class A.

Necessary				Possible			
g_j	Class C	Class B	Class A	g_j	Class C	Class B	Class A
g_1	0.00	40.44	72.88	g_1	0.00	0.01	0.02
g_2	0.20	0.20	0.20	g_2	0.20	660.49	1252.57
g_3	0.00	4.01	6.92	g_3	0.00	7.00	14.42
g_4	17.00	62.40	106.80	g_4	17.00	48.95	74.37

The most balanced development paths in terms of the BP measure for the possible and necessary scenarios are presented in Table 8. When class A should be attained for at least one feasible weight vector, it would be enough to improve on g_2 , g_3 , and g_4 substantially, while not investigated much effort into the most important criterion g_1 . On the contrary, if the target assignment should be reached for all feasible weights, vast improvements need to be made on g_1 , g_3 , and g_4 .

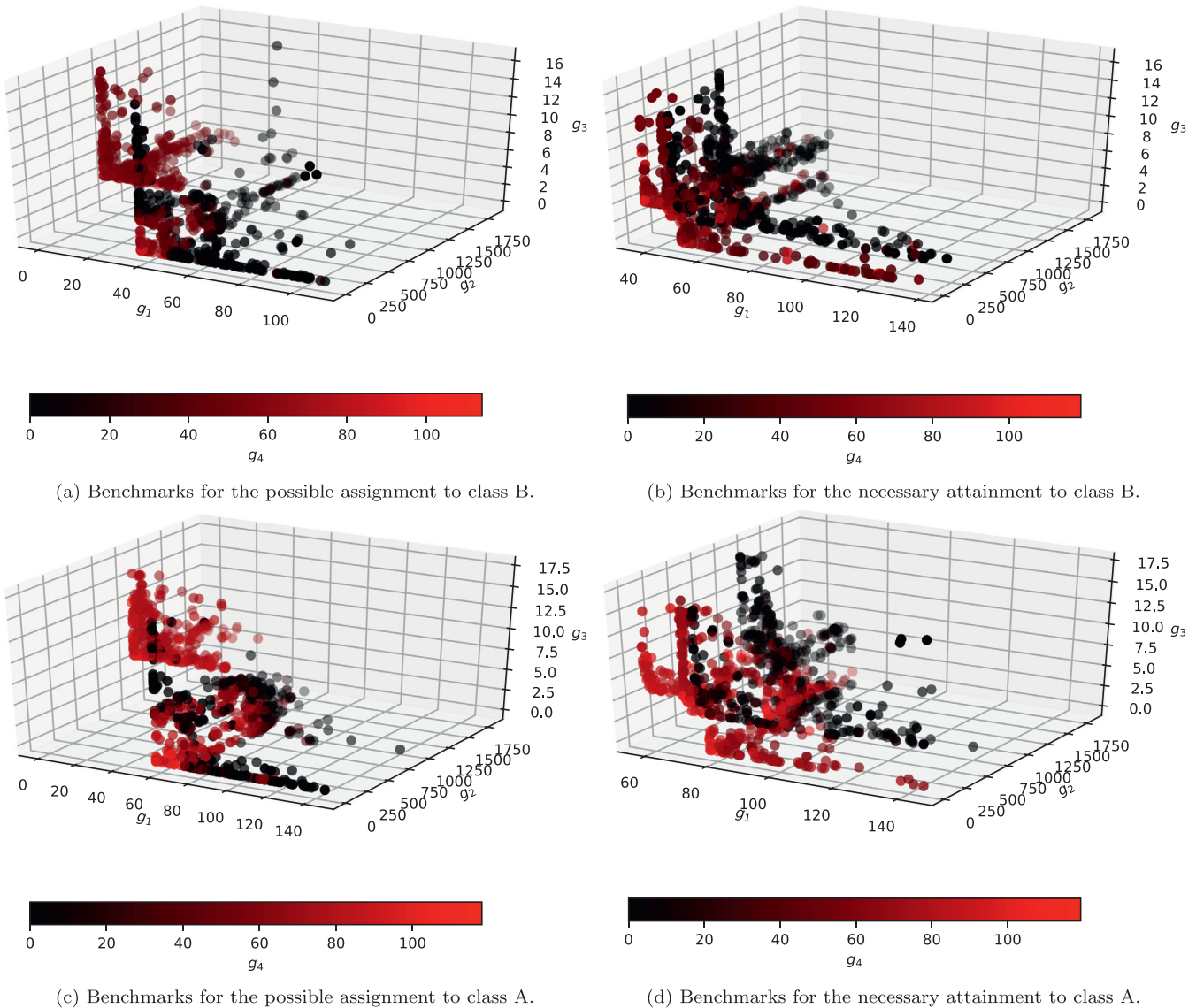


Fig. 6. All possible development paths generated with the EMO-Start-Path algorithm for unit 975 for the possible and necessary assignments to class A (the exact performances of all benchmarks are provided in e-Appendix 2).

5.4. Development paths built on a subset of criteria

In some scenarios, the DMs may indicate a subset of criteria on which the performances should be modified to attain the target assignment, hence leaving aside these criteria on which performances cannot be improved. For illustrative purposes, we aim to possibly assign unit 975 to class A with an intermediate target in class B.

The current performances of unit 975 are low, implying its assignment to the worst class. In particular, it attains the lowest possible ratings on criteria g_1 and g_3 . In this perspective, it is reasonable to analyze which coalitions of criteria would enable unit 975, reaching a more preferred class. In Table 9, we present the example performance improvements on selected subsets of criteria that would allow its possible assignment to class B. Specifically, a substantial modification on the most important criterion g_1 would be sufficient. Even if the reference profile separating classes C and B would be strictly preferred to unit 975 on g_2 , g_3 , and g_4 , the above modification would allow the unit to be weakly preferred over the profile on g_1 . The latter would be sufficient to compare positively to the profile at the comprehensive level.

Table 9

Example performance modifications on selected subsets of criteria that enable unit 975 to possibly attain class B (“-” indicates that the target cannot be achieved when modifying performances only on the respective subset of criteria).

Subset of criteria	g_1	g_2	g_3	g_4
$\{g_1\}$	48.12	0.00	0.00	0.00
$\{g_2\}$	-	-	-	-
$\{g_1, g_3\}$	36.94	0.00	5.27	0.00
$\{g_1, g_4\}$	39.68	0.00	0.00	36.53
$\{g_3, g_4\}$	-	-	-	-
$\{g_1, g_2, g_3\}$	2.03	1.01	17.00	0.00
$\{g_2, g_3, g_4\}$	0.00	690.23	5.87	48.69

When considering the modifications only on g_2 , or g_3 , or g_4 , class B cannot be attained whatever the modification. The same applies to all pairs (e.g., $\{g_3, g_4\}$) composed of the three less important criteria. Thus they need to be combined with g_1 , allowing to decrease the required improvement on this criterion (see, e.g., $\{g_1, g_3\}$ or $\{g_1, g_4\}$ in Table 9). Then, unit 975 is indifferent to the profile on g_1 and strictly preferred on some other criterion (e.g.,

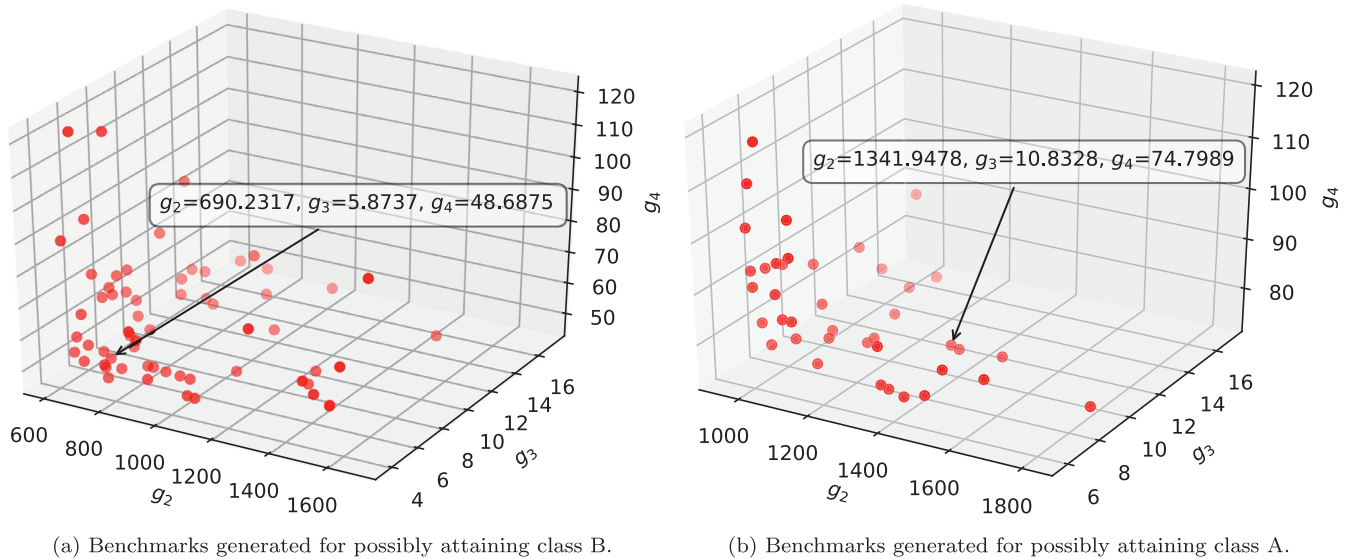


Fig. 7. All development paths generated for unit 975 on a subset of criteria $\{g_2, g_3, g_4\}$ (the exact performances of all benchmarks are provided in e-Appendix 2).

either g_3 or g_4). Alternatively, the performances on all three criteria $\{g_2, g_3, g_4\}$ need to be increased vastly to reach an intermediate target. For the modifications reported in Table 9 for this subset of criteria, unit 975 would be strictly preferred to the profile on g_2 and g_3 and weakly preferred on g_4 .

All two-step development paths based on a subset of criteria $\{g_2, g_3, g_4\}$ are illustrated in Fig. 7. We also mark the best development path in terms of BP built on these criteria. It offers a smooth development plan for the three criteria, indicating that the target assignments can be reached with substantial improvements of the scientific potential and material and non-material effects of the unit's functioning. In real-world decision-making, the generation of development paths may be constrained in terms of a subset of criteria on which the performances can be modified as well as the performance improvements admissible in each step. The proposed method tolerates such constraints. The example results for unit 975 in view of its necessary or possible assignment to class A with an intermediate target in class B are provided in e-Appendix 1.

6. Discussion

6.1. Development planning vs. target setting

Our paper offers an MCDA-based perspective in stepwise benchmarking and introduces a robust model applicable in the context of sorting problems, where the intention to reach the final target is to define and follow the intermediate benchmarks. This creates a development path on route to the best practices or more preferred classes. However, development planning requires not only setting the targets but also investigating how one can reach them.

In the proposed strategies, we considered existing alternatives or fictive benchmarks. In the former case, the underlying logic is that a unit should follow a sequence of relevant benchmarks along its improvement path toward target performance. Doing so allows the DMs to opt for successive improvement and thus set pragmatic aspirations [38]. In other words, the stepwise benchmarking concept is focused on both the target and the question from whom to learn. Investigating the practices and achievements of existing units reported in the survey submitted to the Ministry of Science and Higher Education allows answering how to attain the target performances.

As far as the strategies involving the fictive benchmarks are concerned, one should rather interpret our contribution in terms of target setting. It is a relevant issue in modern literature and the important function of DEA, yielding numerous contributions [7,9,28,29,41,46]. Novel works, including [1], note that DEA-based benchmarking approaches focus precisely on setting the targets while neglecting a question on how to achieve them. When constructing the development paths composed of fictive units, we also identify gradual improvement targets over the relevant criteria. However, a comprehensive development plan should specify how these targets should be reached efficiently and effectively. Such a plan can be formulated for a specific unit after confronting its aspirations with reality. In e-Appendix 1, we enrich the description of the criteria considered in the problem of parametric evaluation with an example interpretation of what means gaining a certain number of points on each criterion given the achievements that are considered when computing the respective performance. For example, the performance on criterion g_1 can be improved by 40 additional points if, e.g., each unit's member would publish a single paper in a journal worth 40 points (e.g., European Journal of Operational Research, Decision Support Systems, or Computers & Industrial Engineering) during four years spanning over the evaluation period. Such observations need to be accounted for when enriching the analysis of the identified targets with a plan on how they should actually be reached. Obviously, this can be done in many different ways on each criterion. Designing algorithms that would exhibit various options remains a future research direction subject to specific research units' interests.

6.2. Static vs. dynamic character of a benchmarking process

Benchmarking has been developed as a practical business tool searching for model solutions to gain the best results [34]. It offers a normative framework around which one can construct a development agenda [35]. The fundamental principles of benchmarking, primarily in a policy-making context, include learning from others, focus, measurement, differentiation, comparability, integration, and applicability. The continuity of a process is also viewed as of great importance [34]. Benchmarking should be a dynamic process that evolves when conditions change. This is a specific setting for the public sector benchmarking and policy-makers.

However, the involvement of dynamics in this process is very problematic as the targeted systems are cumbersome. In partic-

ular, institutional development is a lengthy process. For example, the implementation of the Europe 2020 goals sets the agenda for a decade, and similar is happening with Sustainable Development Goals 2030. A shorter period would undermine laggards to adjust and learn from the benchmarks serving as their role models. Nevertheless, some entities easily reach such fixed benchmarks and accomplish them in a given period. This happened, e.g., in the Europe 2020 strategy, which sets a target for the percentage of the population finishing tertiary education. The target was 40% of the population aged between 30 and 34. Countries such as France and Spain fulfilled the target in a relatively short time span. However, Western Balkan countries and Italy experienced problems reaching the target (e.g., Italy had 27.6% in 2019).

The stepwise benchmarking model can be seen as a static solution that can be revised if policy-makers require updates on the current position. Usually, this is not executed for a period of less than a year. For example, the parametric evaluation of research units is conducted in Poland every four years. The class assignments reflecting the current state and progress are not available in the meantime. Thus, static assumption to create direct benchmarks and route to the final target provides a constructive measurable path to learn, compare, and apply the policy-oriented solutions from benchmarks.

When such a static setting is applied, one runs into the risk that even when the targets are reached, the rating or class assignment of a given unit may not improve in the next evaluation period. For example, if other units improve as well, the authorities may adopt more stringent rules for the evaluation. However, similar problems are experienced in other methods. These include DEA-based approaches that reveal an excess use of some inputs or shortfalls in the production of outputs. Even if these factors are revised by a DMU in line with the formulated recommendations, there is no guarantee that a unit would become efficient when the analysis is repeated after some time. The performances of other DMUs may change too, thus affecting the efficient frontier. Nevertheless, the models incorporating such a static assumption have been found extremely useful and extensively used in practice over the last forty years.

6.3. Generation of a development path directly optimizing a balance measure

This paper introduced the strategies for generating the development paths in a stepwise fashion and a set of measures that can be used to evaluate such paths in terms of the balance of the respective performance modifications. Such a focus can be motivated in a three-fold way. Firstly, we prioritized the minimality of the required performance changes. Thus, we treated them as objectives in the considered optimization problem. When optimizing only the balance of the development path, such minimality would not be guaranteed. As a result, the unit could be recommended a plan composed of well-balanced steps that involve excessive and unnecessary improvements. Secondly, we wished to demonstrate the peculiarities of different measures. For this purpose, we aimed at generating a large set of non-dominated development paths that could be subsequently assessed given various interpretations of their quality. Thirdly, when optimizing the performance changes separately for each step, the number of objectives remained relatively low, not exceeding the number of criteria. In this way, the existing evolutionary methods could efficiently solve the respective multiple objective optimization problems.

However, if a desired feature of the path would be specified a priori, it would be possible to incorporate it as one of the objectives in the optimization problem along with the performance modifications in all steps considered simultaneously. Such a strat-

egy, called *EMO-Ent-Path*, and example results of its use are discussed in e-Appendix 1.

7. Conclusions

We proposed a stepwise benchmarking framework for multiple criteria sorting, where alternatives are assigned to pre-defined and ordered decision classes. Such problems are faced in various areas, including economy, finance, banking, management, environmental risk assessment, resource allocation, electricity market, energy, tourism, and climate change. In many applications, sorting results attained with the current performances serve as a starting point for long-term improvement planning toward reaching higher-level classes. To ease the improvement, the development plan can be decomposed into a few steps of successive performance modifications. The proposed framework finds use in exactly such applications. Its usefulness was illustrated on a problem of parametric evaluation of research units, which is conducted every few years by the Polish Ministry of Science and Higher Education. While referring to a selected group of units representing exact and engineering sciences, we determined the development paths for some example units assigned to the least preferred class. Based on such plans, the management and researchers of a given unit can aim for attaining some pre-defined levels of scientific activity and potential as well as material and non-material effects that would grant the assignment to more preferred, intermediate and target classes.

We proposed strategies for the generation of development paths that incorporate either existing alternatives or fictive benchmarks. A significant advantage of the former approach consists in its realism, i.e., pointing out real alternatives that one should learn from. However, the number of generated development paths in this strategy is limited and influenced by the distribution of alternatives' assignments to different classes. Moreover, the performance modifications suggested in each step may not be minimal. These problems are non-existing when using the fictive benchmarks determined with Post Factum Analysis. On the one hand, their number can be controlled by the parameterization of the evolutionary optimization algorithm. On the other hand, PFA highlights how the alternatives' performances need to be modified minimally to attain the desired sorting recommendation. Nevertheless, the development paths determined with such an approach need to be assessed by the Decision Maker in terms of each step's feasibility. The latter can be supported by indicating a priori both a subset of criteria on which the considered alternative could improve realistically and the maximal admissible performance modifications.

To assess the quality of a plethora of generated development paths, we proposed a set of measures. The basic measures capture the balance in performance modifications while admitting the specification of weights assigned to different criteria and various steps. In this way, we formalized a common interpretation of a good development plan as the one ensuring a gradual improvement of an alternative en route to a more preferred class, with similar advancements within each step. However, we also introduced some supportive measures that quantify the contribution of different criteria in attaining the target assignment.

If the Decision Maker has a good understanding of the desirable characteristics that should be exhibited in the selected path, then a single measure can be used to assess the quality of all constructed development plans. Nonetheless, it is also possible to use a few measures at once to select the path that proves right irrespective of how the concepts of balance or smoothness are interpreted. Alternatively, they can be employed in a lexicographic order so that a subset of good paths in terms of one measure is evaluated with another measure that captures a different perspective on the paths' quality and practical usefulness.

Future research can be oriented to the development or enrichment of different aspects of the proposed method. The original contribution of this paper derives from the focus on multiple criteria sorting problems. However, the introduced stepwise benchmarking approach can also be adjusted to ranking problems. This is particularly appealing within the robustness analysis framework, where the necessary preference relation can be used to determine the possible development paths [18].

The modern benchmarking process may also account for some non-characteristic preferences of Decision Maker set by exogenous boundaries. In particular, (s)he may indicate some specific benchmarks from more preferred classes. The model would find the best development path with the required alternatives as the intermediate benchmarks. For example, in parametric evaluation, one may indicate some research units that work in similar conditions, are close geographically, or are open to cooperation and revealing some of their best practices to others.

When it comes to the measures, they can be revised to consider only a subset of criteria on which the proposed modifications are relatively large. Development paths may suggest marginal changes on some criteria, and in case the modifications in various steps were unbalanced, they would negatively affect the measure values. However, since such criteria do not play an essential role in attaining the target, they may be neglected when evaluating the smoothness of improvements imposed by a given path.

Another interesting direction for future research consists of allowing the Decision Maker to define the importance of each step within the development path. Such importance should not be associated with this step's expected smoothness but rather with how the absolute expected performance modification is related to the changes in other steps. For example, one may require that the initial step modification should be more significant. This is important in policy-based benchmarking as it provides an option for the Decision Maker to control the dynamics within the process. Such a possibility is particularly applicable in long-term benchmarking, such as SDG 2030, where the first step might have more substantial importance due to the necessity for faster results. Closer to this idea are authors in [27], where the computed stepwise benchmarking path depends on the amount of input and output changes allowed in each step.

Finally, the proposed method should be validated in other real-world case studies. The most appealing application areas are in finance [12], banking [21], and business investment [32]. However, this requires a suitable adaptation of the proposed framework to a sorting method used for deriving class assignments in a given study. In this spirit, e-Appendix 1 discusses how the optimization problem presented in Section 3.2 can be adjusted to other popular MCDA sorting methods.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Miłosz Kadziński: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Supervision, Project administration, Funding acquisition. **Mladen Stamenković:** Conceptualization, Methodology, Investigation, Writing – original draft. **Maciej Uniejewski:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – review & editing, Visualization.

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Supplementary material

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